PROJECT PLAN TEMPLATE

Accident Detection and Smart Rescue System

By

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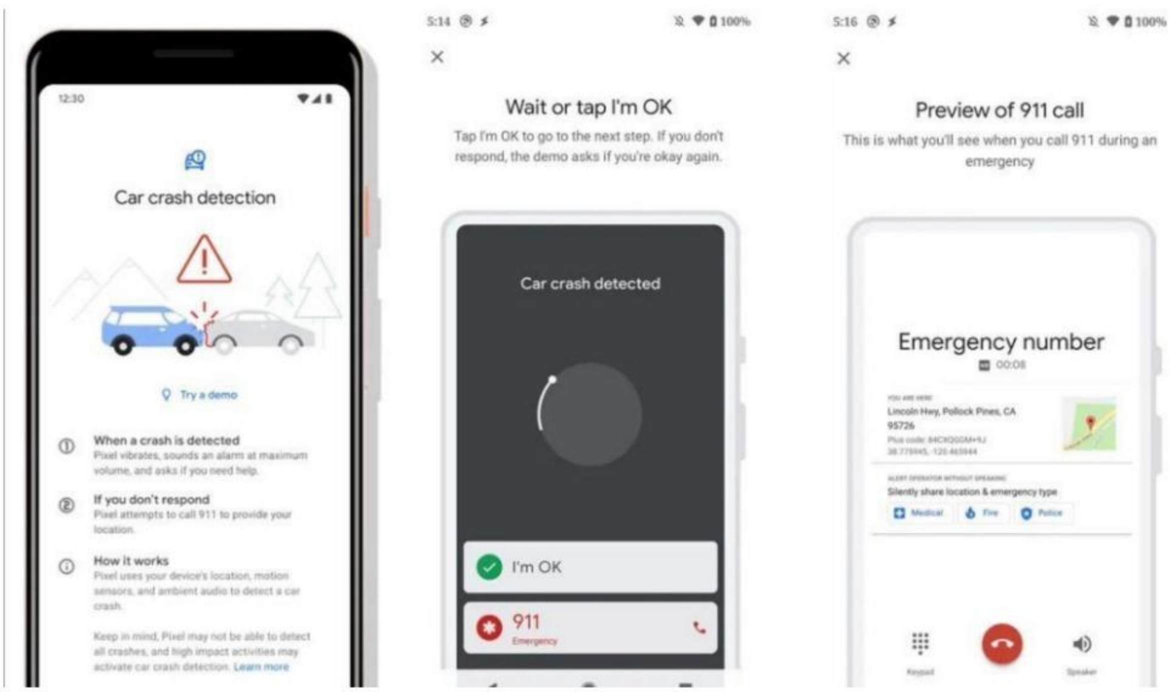
Contents


12:30
Car crash detection
Try a demo
When a crash is detected
Pixel vibrates, sounds an alarm at maximum volume, and asks if you need help.
If you don't respond
Pixel attempts to call 911 to provide your location.
How it works
Pixel uses your device's location, motion sensors, and ambient audio to detect a car crash.
Keep in mind, Pixel may not be able to detect all crashes, and high impact activities may activate car crash detection. Learn more
5:14
100%
Wait or tap I'm OK
Tap I'm OK to go to the next step. If you don't respond, the demo asks if you're okay again.
Car crash detected
I'm OK
911
Emergency
5:16
100%
Preview of 911 call
This is what you'll see when you call 911 during an emergency
Emergency number
00:08
Lincoln Hwy, Pollock Pines, CA 95726
Silently share location & emergency type

Section 1. Project Overview

1.1 Project Description

- A large number of deaths and disability cases are caused by Traffic accidents every day. Most of fatalities occur because of the late response to these emergency cases. In many situations the family members or emergency services aren't informed in time which causes death or severe injury. The time after the traumatic injury is called the golden hour, where providing essential medical and surgical aid at that time increases the probability of saving human lives by one third an average.
- There are many solutions proposed to avoid the problem. We have avoided as well as some extent but still we can't avoid it completely. Many lives are lost due to improper post accident signaling and tracing out the exact location.
- Our project provides solution for the above stated problem which involves intimating the nearby hospitals and relatives by giving the accident location of vehicle using GSM and GPS technologies. Heart of our project is the android app where we can use the existing internal hardware modules like GPS, GSM etc. to get/send the required information to the concerned persons. This way we can also minimize the project cost as well. To come over this we are going to design android application that detects and classifies vehicle accidents based on severity level and reports the essential information about the accident to emergency services providers. So, we can save time to rescue people lives.

1.2 Project Scope

The system includes a sensor, sound meter, GPS and GSM. The sensors will detect the accident, sound meters will trigger an alarm. The GPS will track the location coordinates and the GSM will send an alert notification to the nearby hospital and police, to notify them of the accident occurred to take immediate actions.

Project Includes
Immediate sending to the ambulance unit, police and if there isn’t respond then a call is made using GSM in the smart phone.
Availability of canceling fault alert as system gives the user only 5 seconds to canceling the alert in case it was wrong detecting.
Includes user information such as address, one of his family members’ number.
Server Database to Notifying and Tracking Ambulance.
The support of sensors like vibration sensor, MEMS (micro electrical mechanical system), GPS and GSM (Global System for Mobile communications).
User medical information about health in case he/she needed to enter the hospital.

Project Excludes
No filter in place to verify if sensor detected wrong or not.
responders need to check the web server for accident notification.
The availability of the ambulances.
The availability of a place in the hospital.
If unfortunately the accident is deadly and the user died.
If the phone battery is dead by any means, then we can’t intimate to the concerned people.

1.3 Assumptions

Assumptions
Availability at any time.
Application is easy to use not complicated.
User must have smart phone with needed sensors.
The smart phone must be connected to the internet while in the car.
Hospital or ambulance unit has the ability to respond.
The user must specify all the information to be correct to determine the patient's condition to speed up the rescue process.

1.4 Constraints

Constraints
Budget constraints:Total project cost should not exceed $25,000.
Time constraints:Completion within 7 months.
Loss of internet connection.
Any damage to the smart phone.
Low awareness of technologies.
User entering invalid data.

Section 2. Project Start-Up

2.1 Project Life Cycle:

Phase	Step
Planning:	• Outline the scope of the problem and identify solutions. Resources, costs, time. • It also collects as much as possible details about the system. • Identify high level detailed work plan for the selected project and strategy how to be implemented. • Helps in determine the project goals. • Calculating labor and material costs. • Creating a timetable with target goals. • Creating the project's teams and leadership structure.
Analysis:	• Break down the deliveraebles in the high level Project charter into the more detailed business requirements. • Identify the overall direction that the project will take through the creation of the project strategy documents. • Gathering Requirements which consider the main attraction of this phase. • Define any prototype system required for the system. • In this phase, the requirement gathered in the SRS document (Software requirements specification).

Design:	● Requirements are used to create specifications of the required system. ● Developers will first outline the details for the overall application, such as its: user interfaces, Databases. ● Design and integrate database. ● Design the user interfaces. ● Design integrate system controls
Implementation:	● putting the project plan into action. ● Project developers begin building and coding the software. ● Made the Documentation and training materials. ● Database admins create the necessary data in the database. ● Front-end developers create the necessary interfaces and GUI. ● Build the UI Designs and implement the software functions. ● Verify and test. ● train the staff in the system. ● Install the system

2.2 Methods, Tools, and Techniques:

Methods	• **Software Management:** (Data Collection, Agile Development, Risk Analysis). • **Software Requirements:** (Requirements Specification). • **Software Design:** (Database Management System (DBMS)). • **Software Construction:** (Detailed Design- Code Review).
Programming language	• Using Dart language and Flutter to make the application. • Using MySQL to create a database. • Python, Java, PHP, SQL for backend.
Tools & Techniques	• **IDES:** Android Studio / visual studio code / MySQL Workbench. • **Diagrams:** Network diagram / Gantt chart / ERD / DFD • Adobe XD / Dia Diagram Editor

2.3 Estimation Methods and Estimates:

1. The project has many factors to be completed, like making the application requirements, planning and investigation, system analysis and design, creating database.
2. Getting the database realized using MySQL, designing the project front-end, implementing the actual application in back-end.
3. Implementation by using JavaScript and python.
4. Testing is very important to handle any error occurrence, and finally creating the application.

Estimation Methods and Estimates			
Description	**Best**	**Most Likely**	**Worst**
Effort in person months or person-hours	20 days	25 days	30 days
Schedule in calendar months			
Determine purpose and goals	1 month	45 days	2 month
Planning initiation	4 weeks	5 weeks	6 weeks
Resource planning	3 weeks	4 weeks	5 weeks
Mapping	10 days	15 days	20 days
Determine target	1 days	3 days	5 days
Front end design	2 weeks	3 weeks	4 weeks
Database design	1 weeks	2 weeks	3 weeks
Back-end programming	1 weeks	2 weeks	3 weeks
Testing	2 days	4 days	6 days
Launch	1 day	2 days	3 days
Budget in dollars	$70.000	$80.000	$100.000
Level of Uncertainty	25%	15%	40%
employee salary	Every week	Half a month	Every month

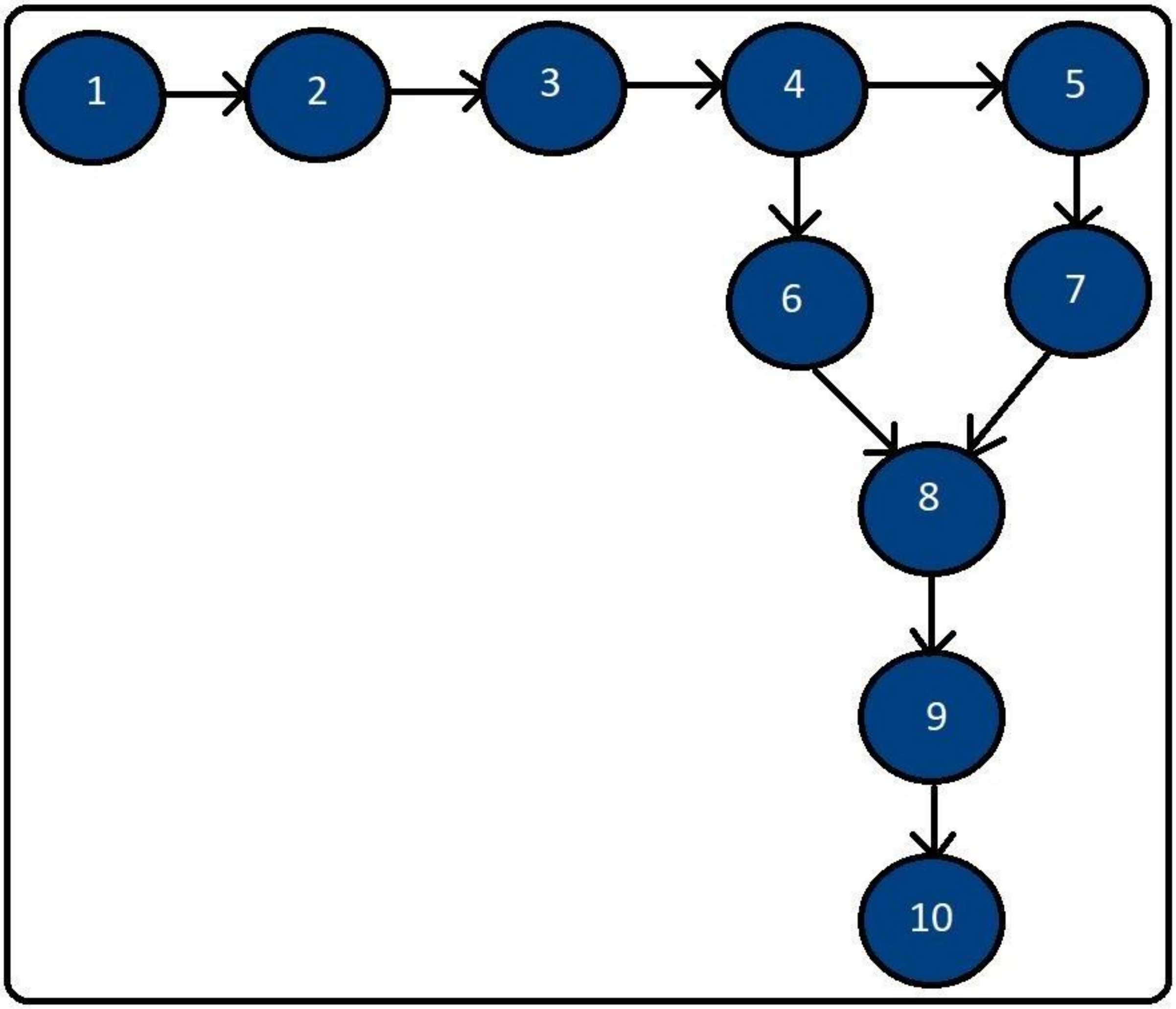
2.4 Schedule Allocation:

Activities	Duration
Determine Purpose and Goals	45 Days
Planning Initiation	35 Days
Resource Planning	28 Days
Mapping	15 Days
Determine Target	3 Days
Front end Design	21 Days
Database Design	14 Days
Back-end Programming	14 Days
Testing	4 Days
Launch	2 Days

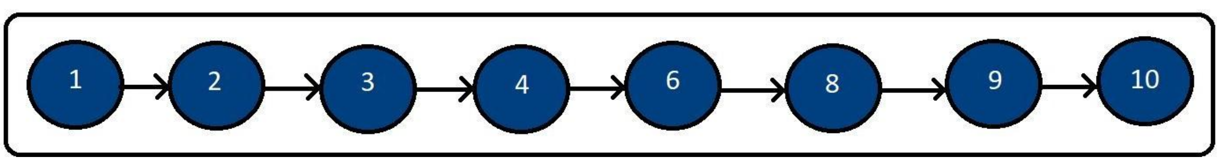
2.5 Network Diagram:

Number	Tasks	Pre-task	ET	TE	TL	Slack Time	Critical Path
1	Determine Purpose and Goals	-	45 days	45	45	0	yes
2	Planning Initiation	1	35 days	80	80	0	yes
3	Resource Planning	2	28 days	108	108	0	yes
4	Determine Target	3	15 days	123	123	0	yes
5	Mapping	4	3 days	126	130	4	-
6	Front end Design	4	21 days	144	144	0	yes
7	Database Design	5	14 days	140	144	4	-
8	Back end Programmi ng	6,7	14 days	158	158	0	yes
9	Testing	8	4 days	162	162	0	yes
10	Launch	9	2 days	164	164	0	yes

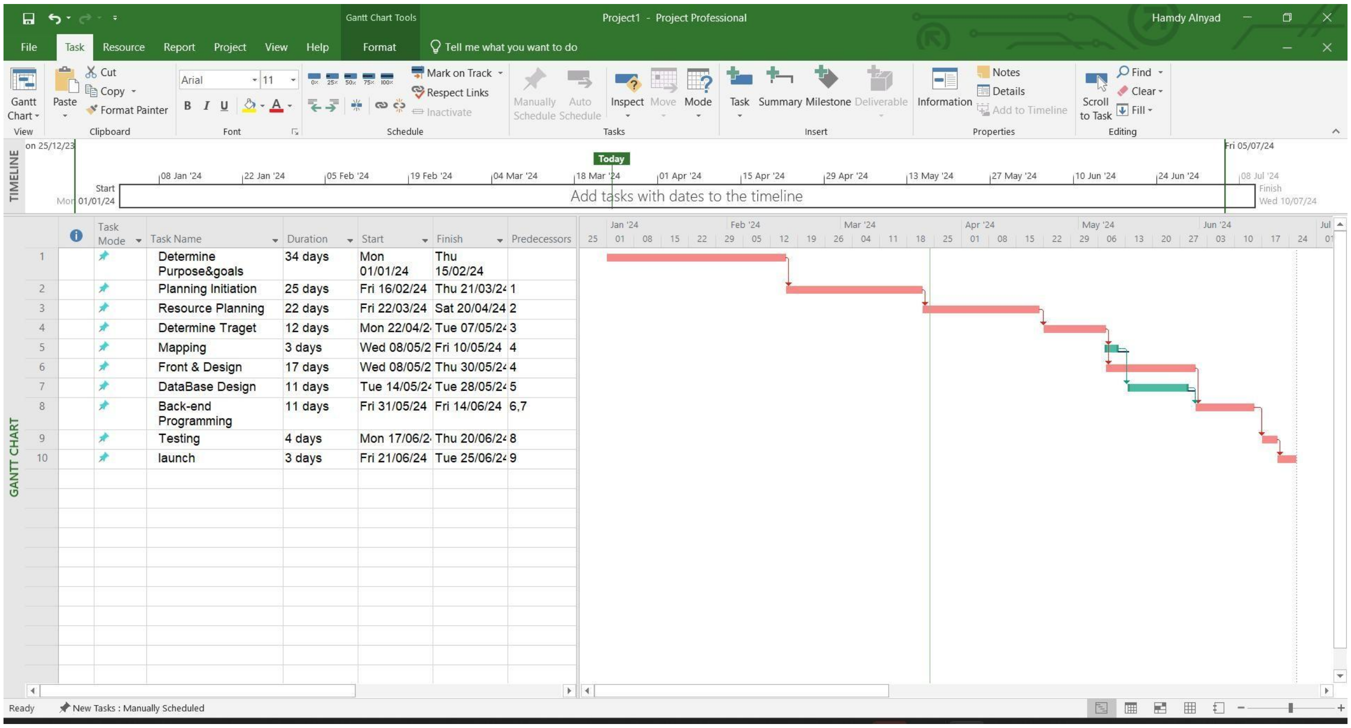
Activities	Start Date	End Date
Determine Purpose and Goals	1 /1/ 24	2 /15/24
Planning Initiation	2 /16/24	3 /21/24
Resource Planning	3 /22/24	4/ 20/24
Determine Target	4 /21/24	5/6/24
Mapping	5/7/24	5/10/24
Front end Design	5/11/24	6/2/24
Database Design	6/3/24	6/17/24
Back-end Programming	6/18/24	7/2/24
Testing	7/ 3/24	7/ 7/24
Launch	7/8/24	7/10/24



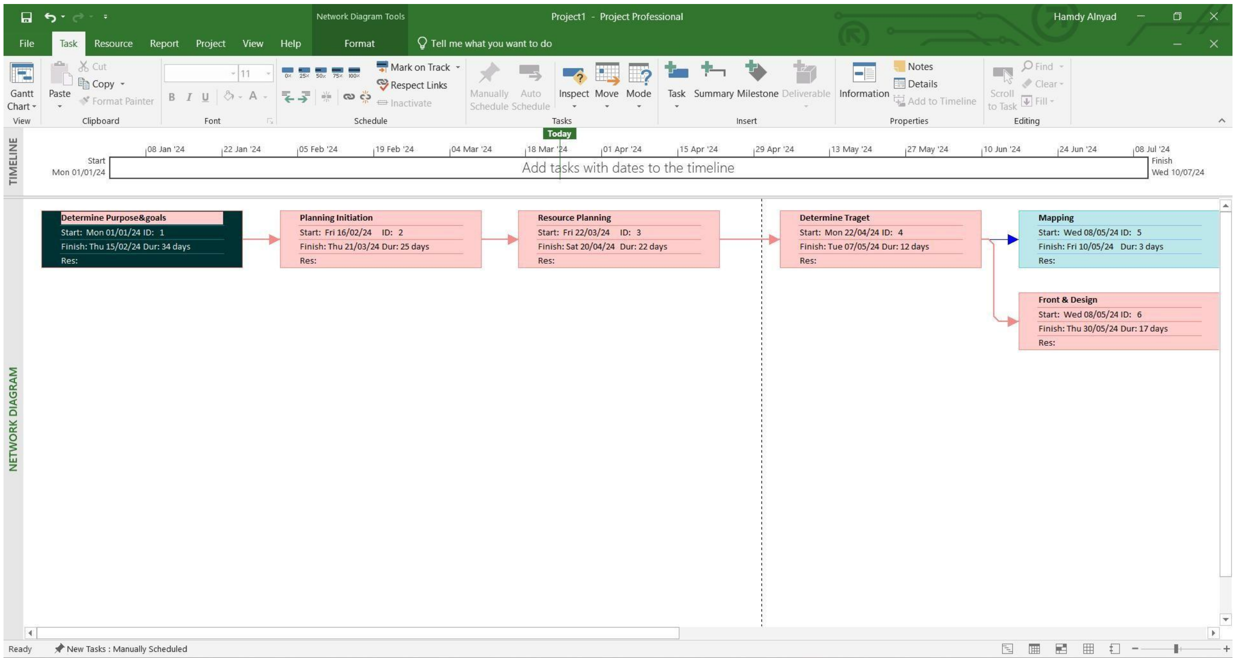
Critical Path:

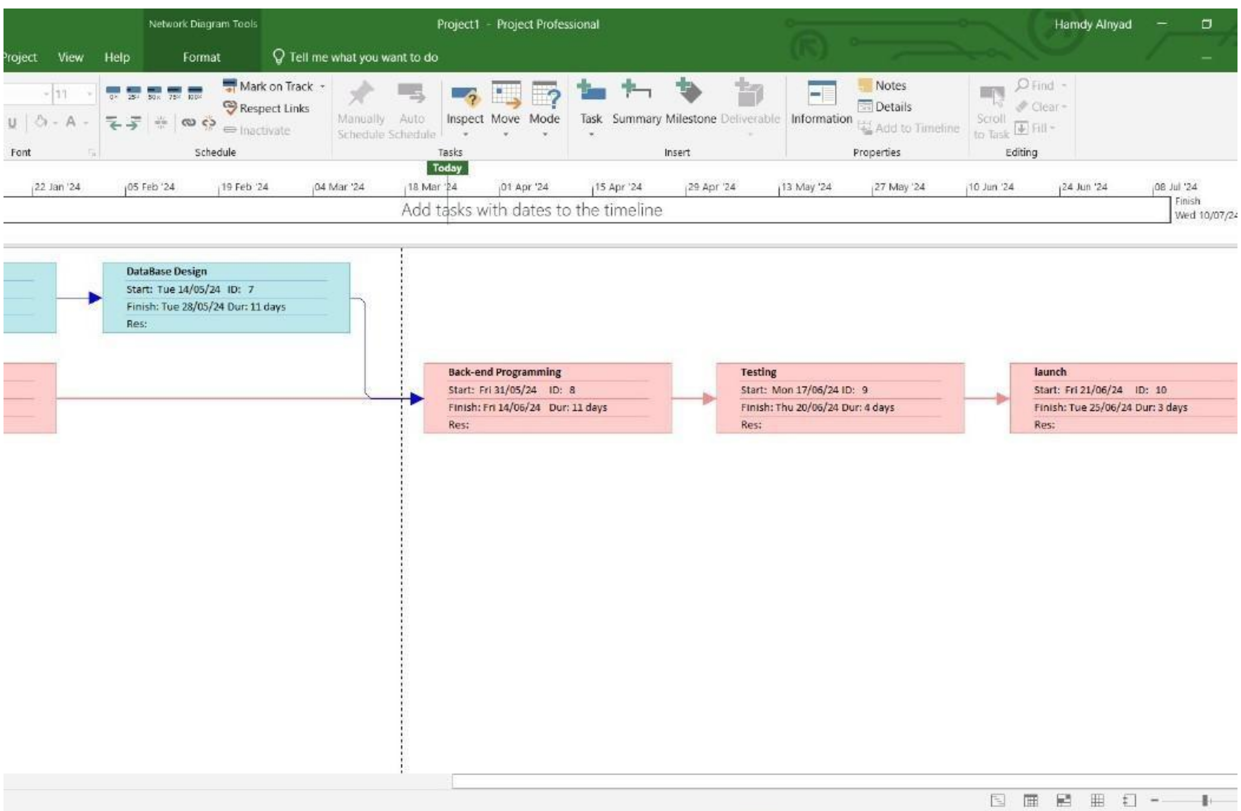


Gantt Chart:



	Task Name	Duration	Start	Finish	Predecessors
1	Determine Purpose&goals	34 days	Mon 01/01/24	Thu 15/02/24	
2	Planning Initiation	25 days	Fri 16/02/24	Thu 21/03/2	1
3	Resource Planning	22 days	Fri 22/03/24	Sat 20/04/24	2
4	Determine Traget	12 days	Mon 22/04/2	Tue 07/05/2	3
5	Mapping	3 days	Wed 08/05/2	Fri 10/05/24	4
6	Front & Design	17 days	Wed 08/05/2	Thu 30/05/2	4
7	DataBase Design	11 days	Tue 14/05/2	Tue 28/05/2	5
8	Back-end Programming	11 days	Fri 31/05/24	Fri 14/06/24	6,7
9	Testing	4 days	Mon 17/06/2	Thu 20/06/2	8
10	launch	3 days	Fri 21/06/24	Tue 25/06/2	9


Project1 - Project Professional
Network Diagram Tools
Hamdy Alnyad
Add tasks with dates to the timeline
Determine Purpose&goals
Start: Mon 01/01/24 ID: 1
Finish: Thu 15/02/24 Dur: 34 days
Res:
Planning Initiation
Start: Fri 16/02/24 ID: 2
Finish: Thu 21/03/24 Dur: 25 days
Res:
Resource Planning
Start: Fri 22/03/24 ID: 3
Finish: Sat 20/04/24 Dur: 22 days
Res:
Determine Traget
Start: Mon 22/04/24 ID: 4
Finish: Tue 07/05/24 Dur: 12 days
Res:
Mapping
Start: Wed 08/05/24 ID: 5
Finish: Fri 10/05/24 Dur: 3 days
Res:
Front & Design
Start: Wed 08/05/24 ID: 6
Finish: Thu 30/05/24 Dur: 17 days
Res:


Project1 - Project Professional
Network Diagram Tools
Hamdy Alnyad
Add tasks with dates to the timeline
DataBase Design
Start: Tue 14/05/24 ID: 7
Finish: Tue 28/05/24 Dur: 11 days
Res:
Back-end Programming
Start: Fri 31/05/24 ID: 8
Finish: Fri 14/06/24 Dur: 11 days
Res:
Testing
Start: Mon 17/06/24 ID: 9
Finish: Thu 20/06/24 Dur: 4 days
Res:
launch
Start: Fri 21/06/24 ID: 10
Finish: Tue 25/06/24 Dur: 3 days
Res:

2.6 Feasibility Study:

-Benefit Per Year = 150,000$

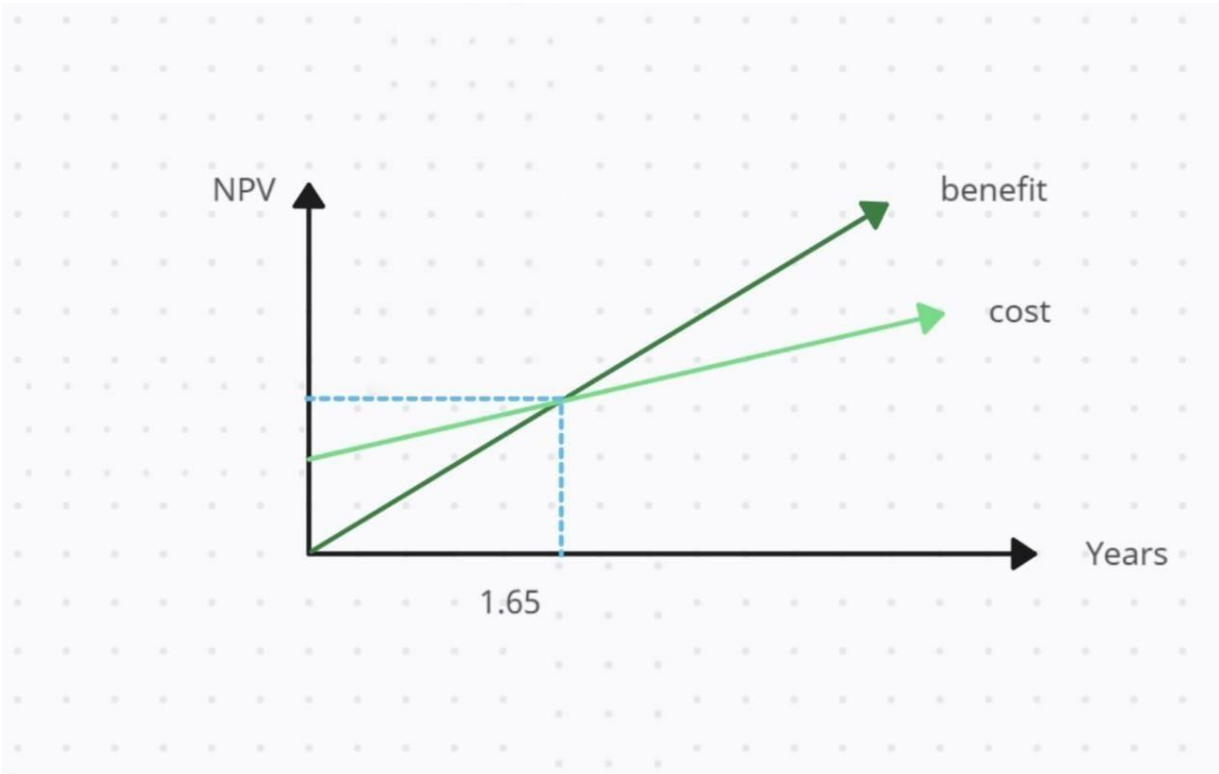
-Discount Rate = 10%

-Recurring Cost Per Year = 60,000$

-One Time Cost = 130,000$

	Year of Project						
	Year_0	Year_1	Year_2	Year_3	Year_4	Year_5	Total
Net economic benefit	0$	150.000$	150.000$	150.000$	150.000$	150.000$	-
Discount Rate (10%)	1	0.9090	0.8264	0.7513	0.6830	0.6209	-
PV of Benefits	0$	136,364$	123,967$	112,697$	102,452$	93,138$	-
NPV of all Benefits	0$	136,364$	260,331$	373,028$	475,480$	568,618$	568,618$
One Time Costs	(130,000$)						
Net economic Cost	0$	60,000$	60,000$	60,000$	60,000$	60,000$	60,000$
Discount Rate (10%)	1	0.9090	0.8264	0.7513	0.6830	0.6209	-
PV of Cost	0$	54,540$	49,584$	45,078$	40,980$	37,254$	-
NPV of all Cost	(130,000$)	184,540$	234,124$	279,202$	320,182$	357,436$	357,436$
Overall NPV	211,182$						
Overall ROI	0.5908						
Yearly PV Cash Flow	(130,000$)	81,824$	74,383$	67,619$	61,472$	55,884$	-
Yearly Overall NPV Cash Flow	(130,000$)	48,176$	26,207$	93,826$	155,298$	211,182$	-
Break Even Point	0.65						
Break Point Occurs	1.65						

Break Point Even Occurs:



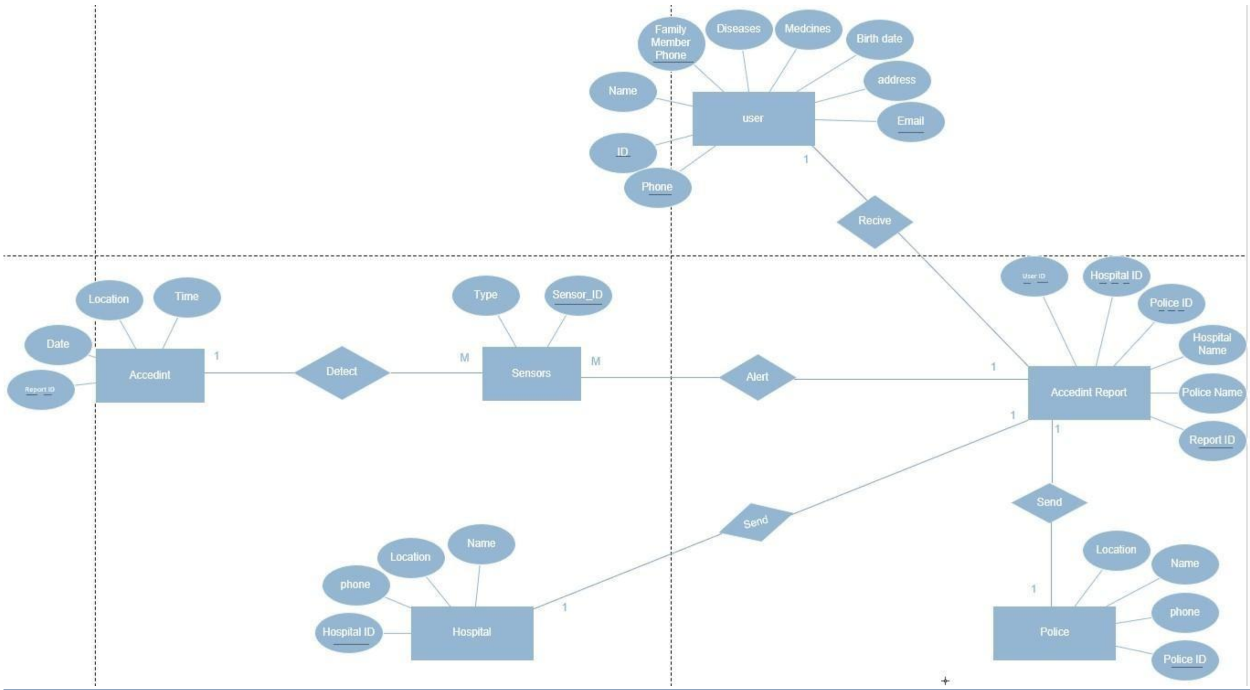
2.7 Resource Allocation:

Resource	Total	Skill Set Requirements	Time frame
Data Collector	2	Search and collect data	45 days
UI designer	1	Adobe XD	5 days
Front-end developer	2	Flutter	3 weeks
Back-end developer	3	Python, Java, PHP, SQL	2 weeks
Database designer	3	DML, DCL	2 weeks

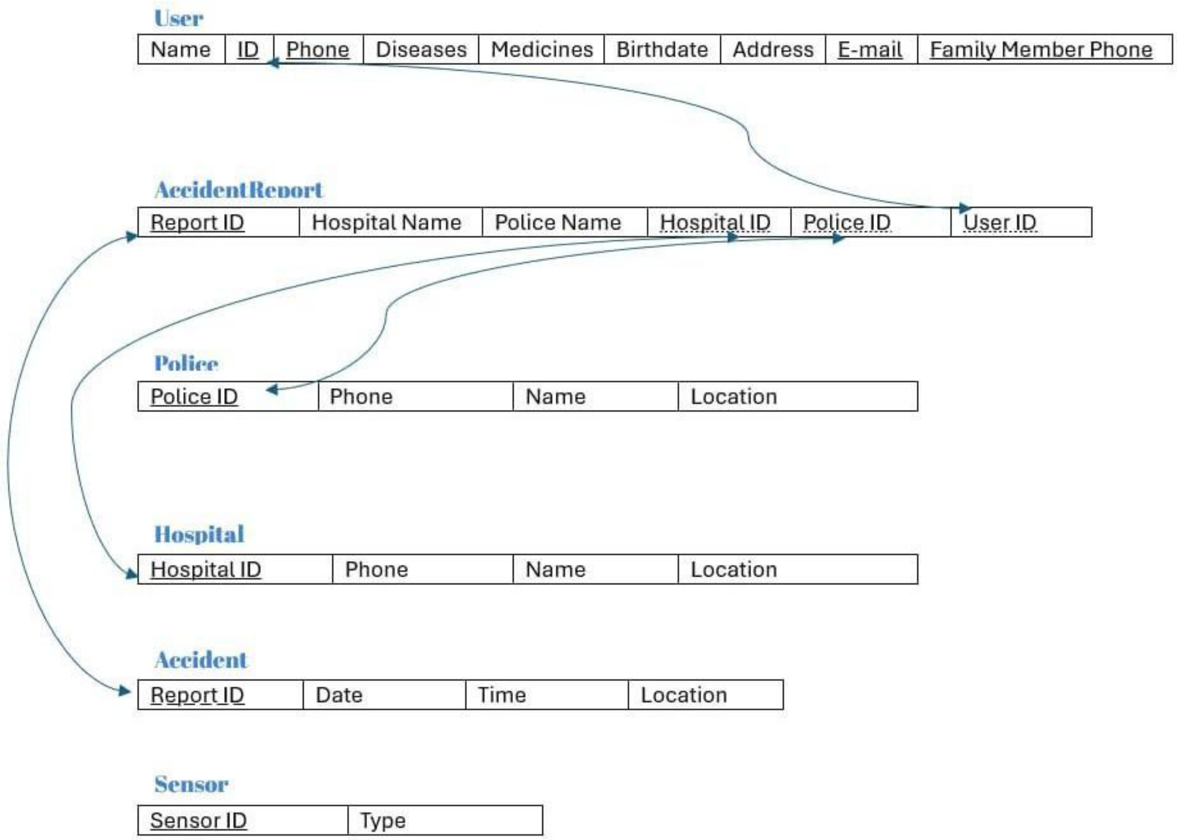
2.8 Budget Allocation:

Key Budget Category	Budget Amount
Determine purpose and goals	$3,000
Planning initiation	$5,000
Resource planning	$30,000
Mapping	$12,000
Front end design	$15,000
Database design	$10,000
Back-end programming	$25,000
Testing	$5,000
Launch	$2,000

2.9 ERD:


Family Member Phone
Diseases
Medcines
Birth date
Name
address
user
Email
ID
Phone
1
Recive
Location
Time
Type
Sensor_ID
User ID
Hospital ID
Police ID
Date
Hospital Name
Report ID
Accedint
1
Detect
M
Sensors
M
Alert
1
Accedint Report
Police Name
1
1
Report ID
Send
Send
Location
Name
Location
Name
phone
1
phone
1
Hospital ID
Hospital
Police
Police ID

2.10 Mapping:


User
Name | ID | Phone | Diseases | Medicines | Birthdate | Address | E-mail | Family Member Phone
AccidentReport
Report ID | Hospital Name | Police Name | Hospital ID | Police ID | User ID
Police
Police ID | Phone | Name | Location
Hospital
Hospital ID | Phone | Name | Location
Accident
Report ID | Date | Time | Location
Sensor
Sensor ID | Type

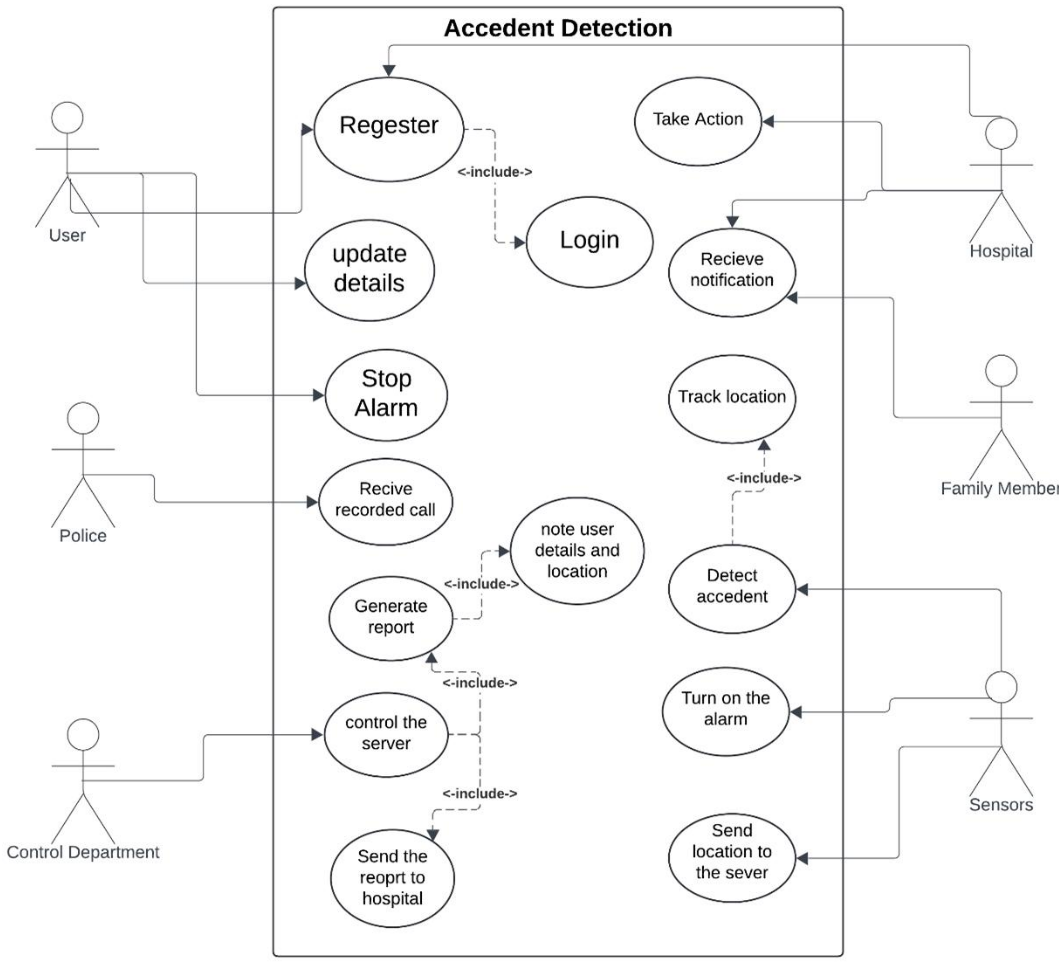
Section 3. Risk Management

Risk Description	Probability	Impact	Strategy
Cost Estimates Unrealistic	Low	High	Included in project plan, subject to amendment as new details regarding project scope are revealed
Time Estimates Unrealistic	Medium	High	Always Mention the Worst Cases
Team Size	Low	Low	Try to Fit Tasks with the available Number of Team Members
Project Scope Creep	Low	High	Defined in project plan, reviewed by Project Manager and Steering Committee to prevent scope creep
Team Members unknowledgeable of Business	Low	Medium	Get Information Throughout searching and provide Them with Documentations
Narrow Knowledge Level of Users	High	Medium	Teaching Them How to Get Information Throughout searching and provide Them with Documentations && Manual use
Server Hacking	Low	High	Start Working on Backup Data till Fix the Error and Protect the System
Programmer Suddenly Absent	Low	Medium	Working With the Available Number of Team Members
Communication failure	Medium	High	Implement backup communication channels and conduct regular testing to verify the reliability of communication systems
Inaccurate GPS Coordinates	Medium	High	Implement error-checking mechanisms and integrate multiple location-detection technologies for more accurate results.
Integration Challenges	Low	High	Collaborate with relevant authorities and agencies, and conduct thorough

			compatibility tests during the integration phase.

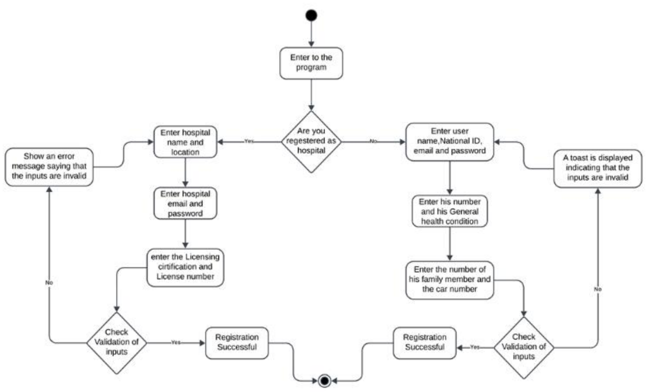
Section 4. Project Design

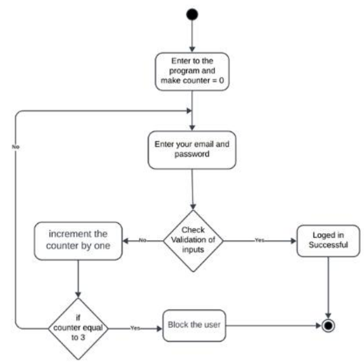
4.1 Use Case:



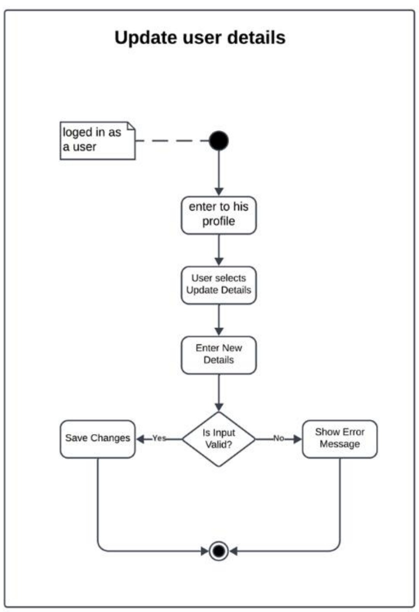
4.2 Activity Diagram:

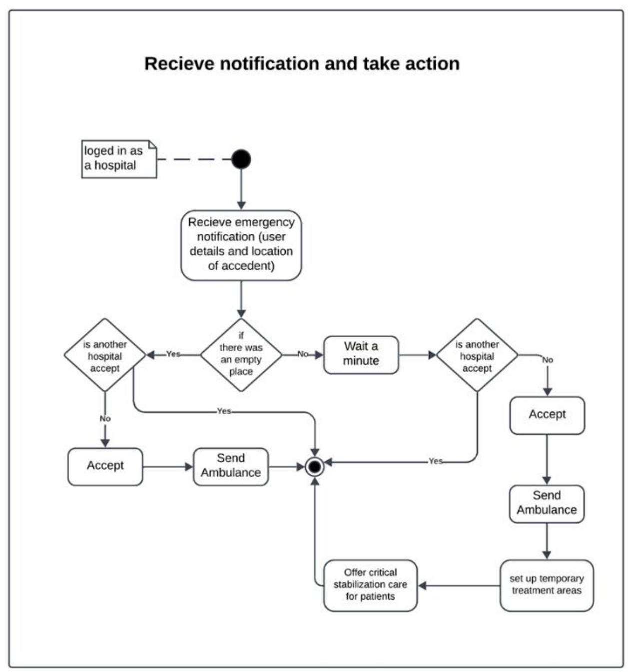
4.2.1: Register & Login:



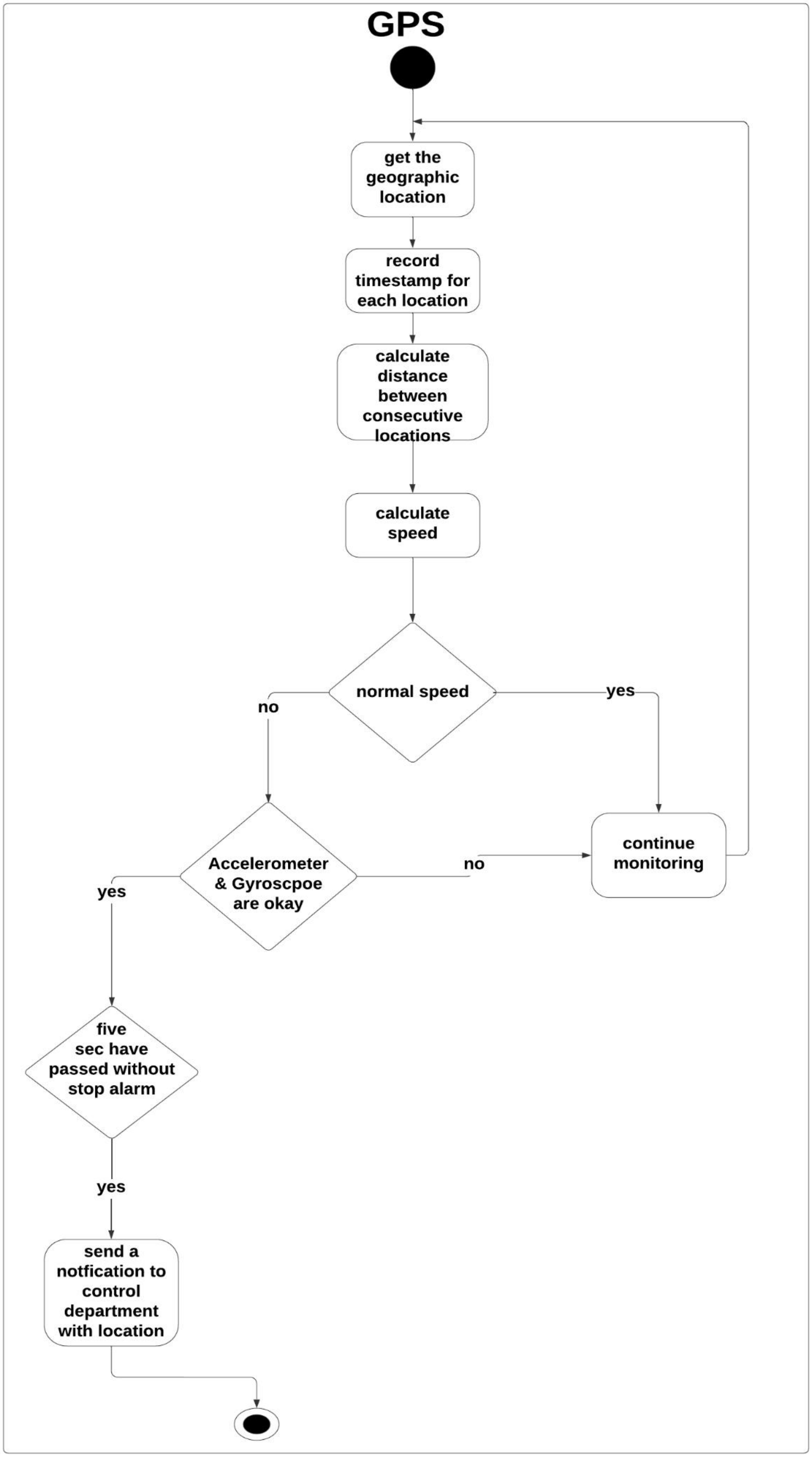


4.2.2: update:

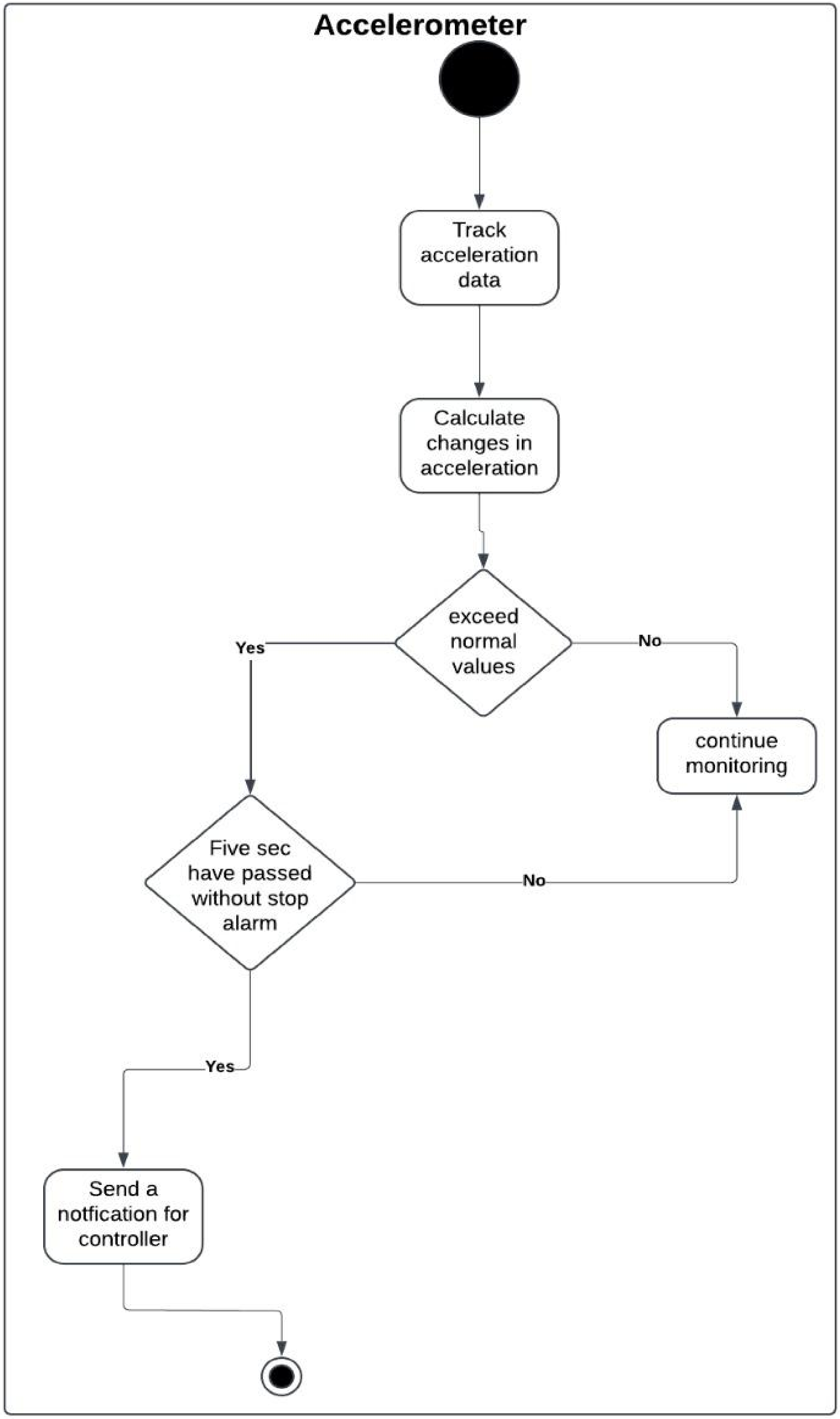




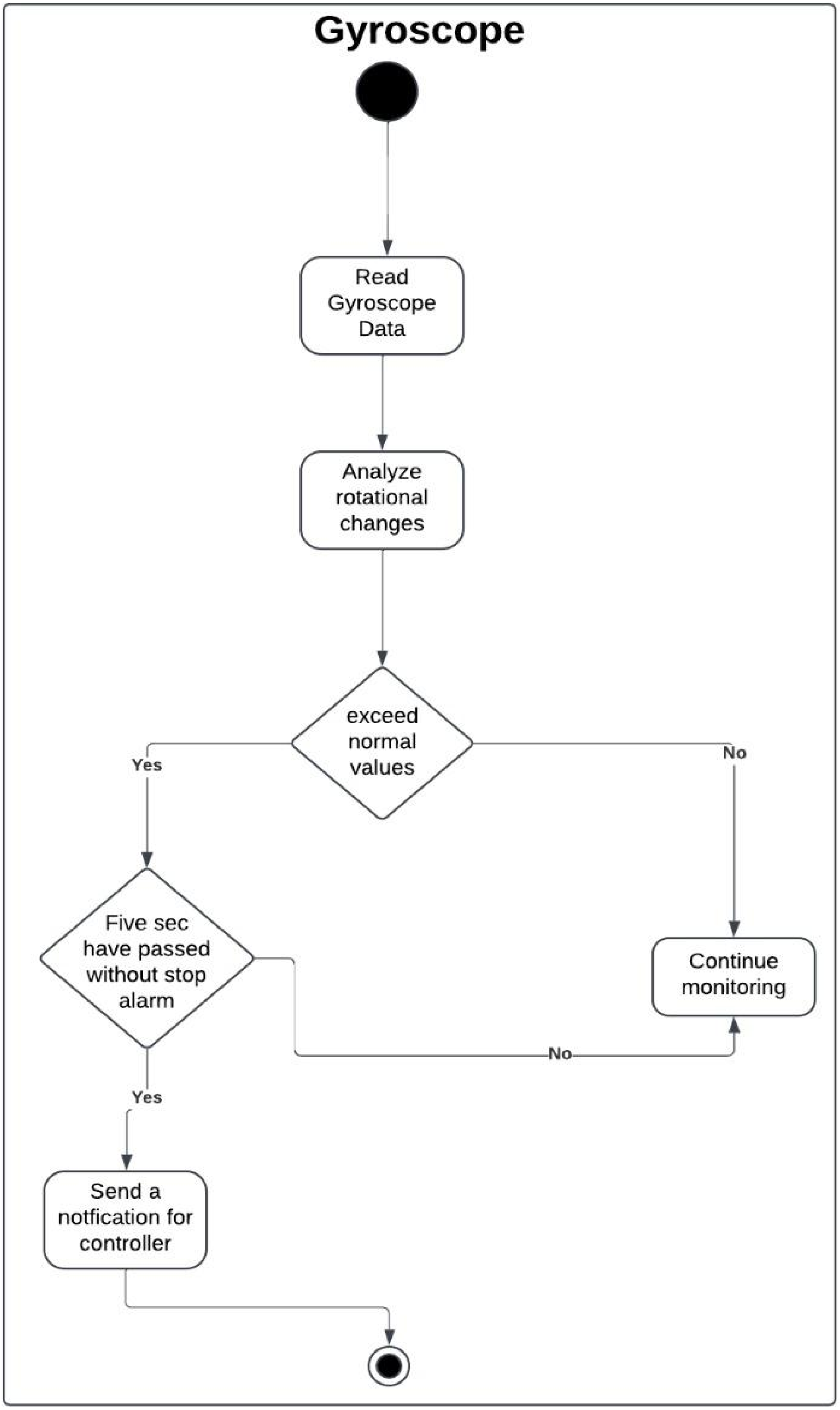
4.2.3: GPS


GPS
get the geographic location
record timestamp for each location
calculate distance between consecutive locations
calculate speed
normal speed
no
yes
Accelerometer & Gyroscpoe are okay
no
continue monitoring
yes
five sec have passed without stop alarm
yes
send a notfication to control department with location

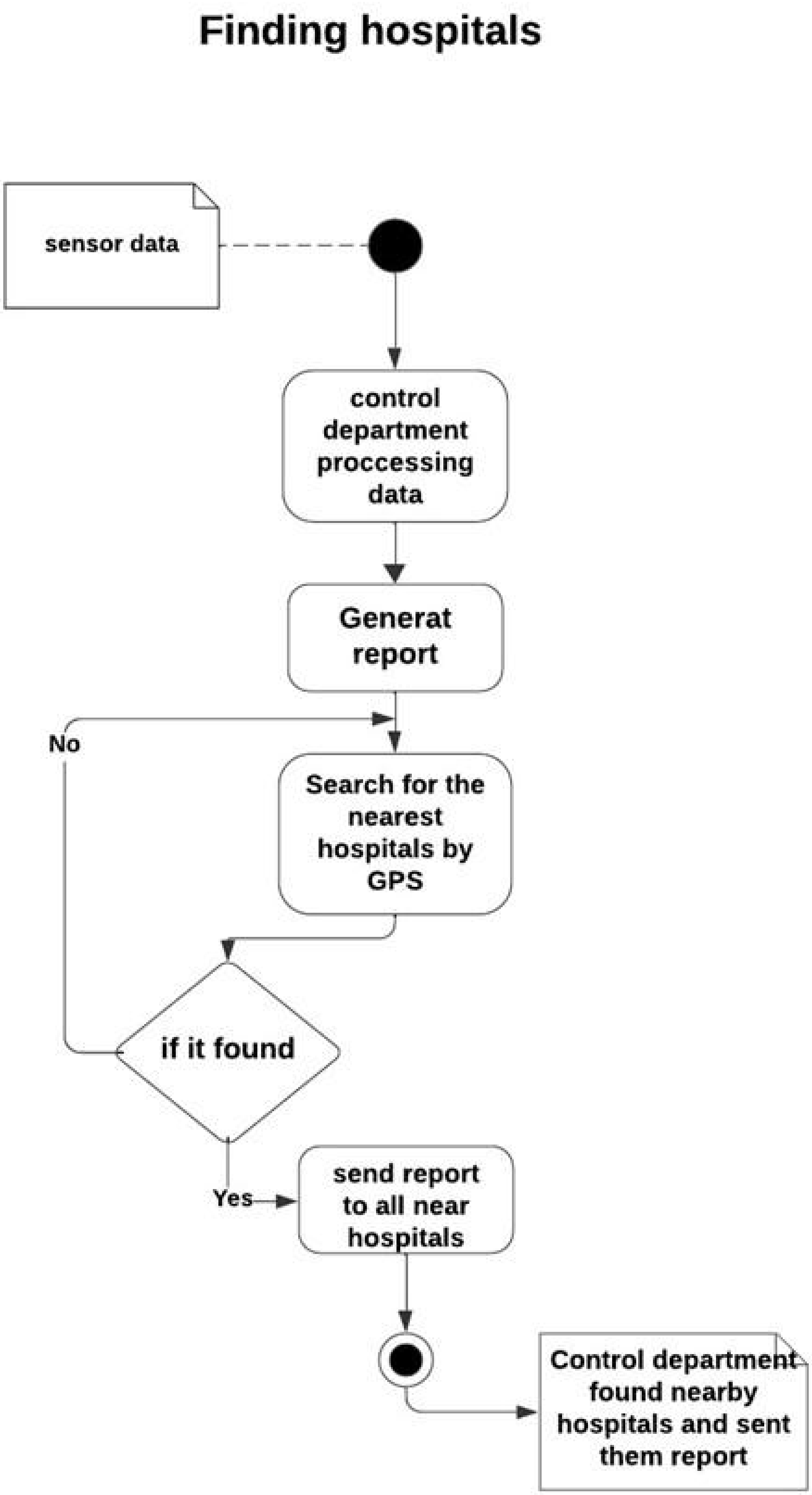
4.2.4: Accelerometer



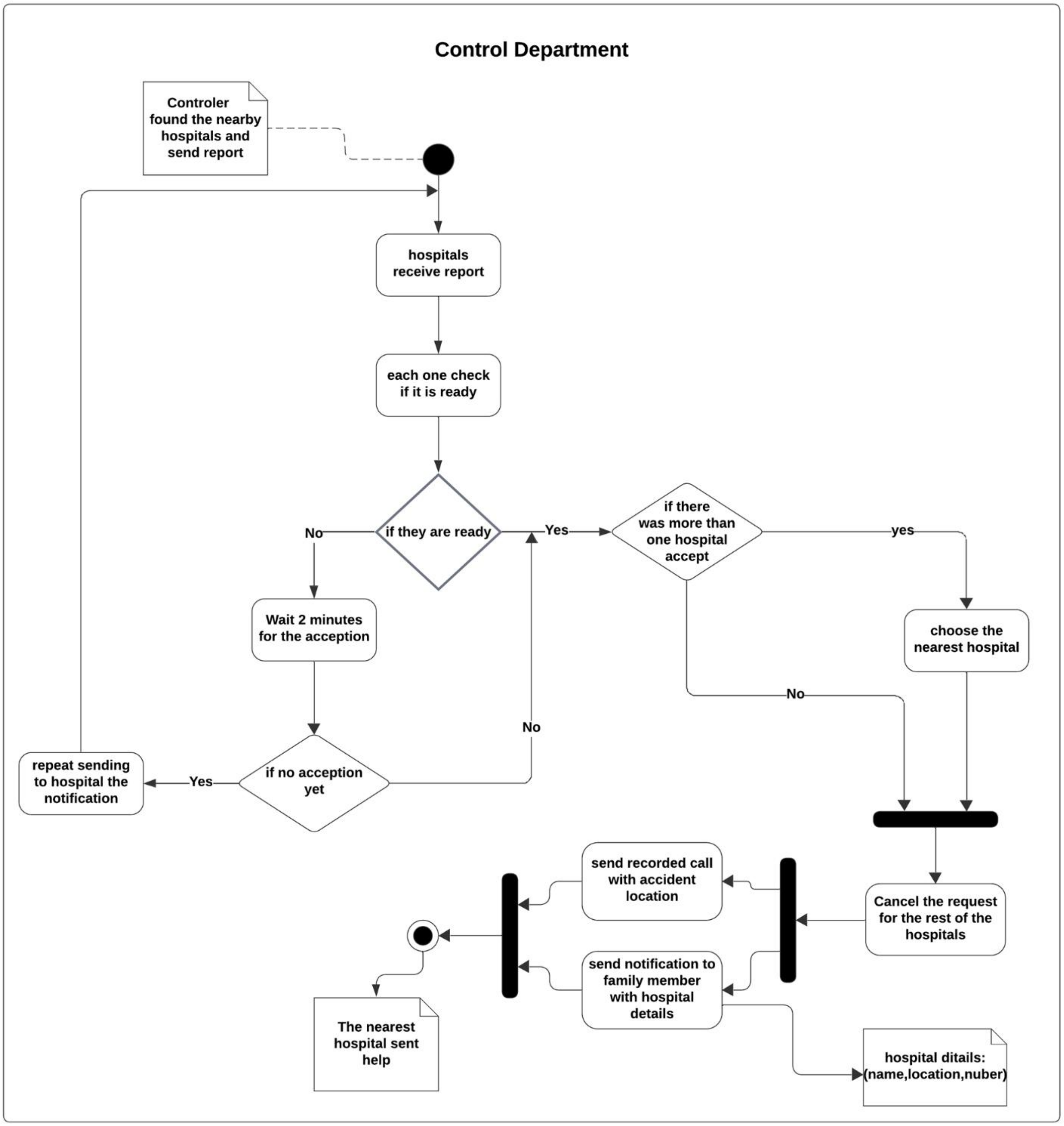
4.2.5: Gyroscope



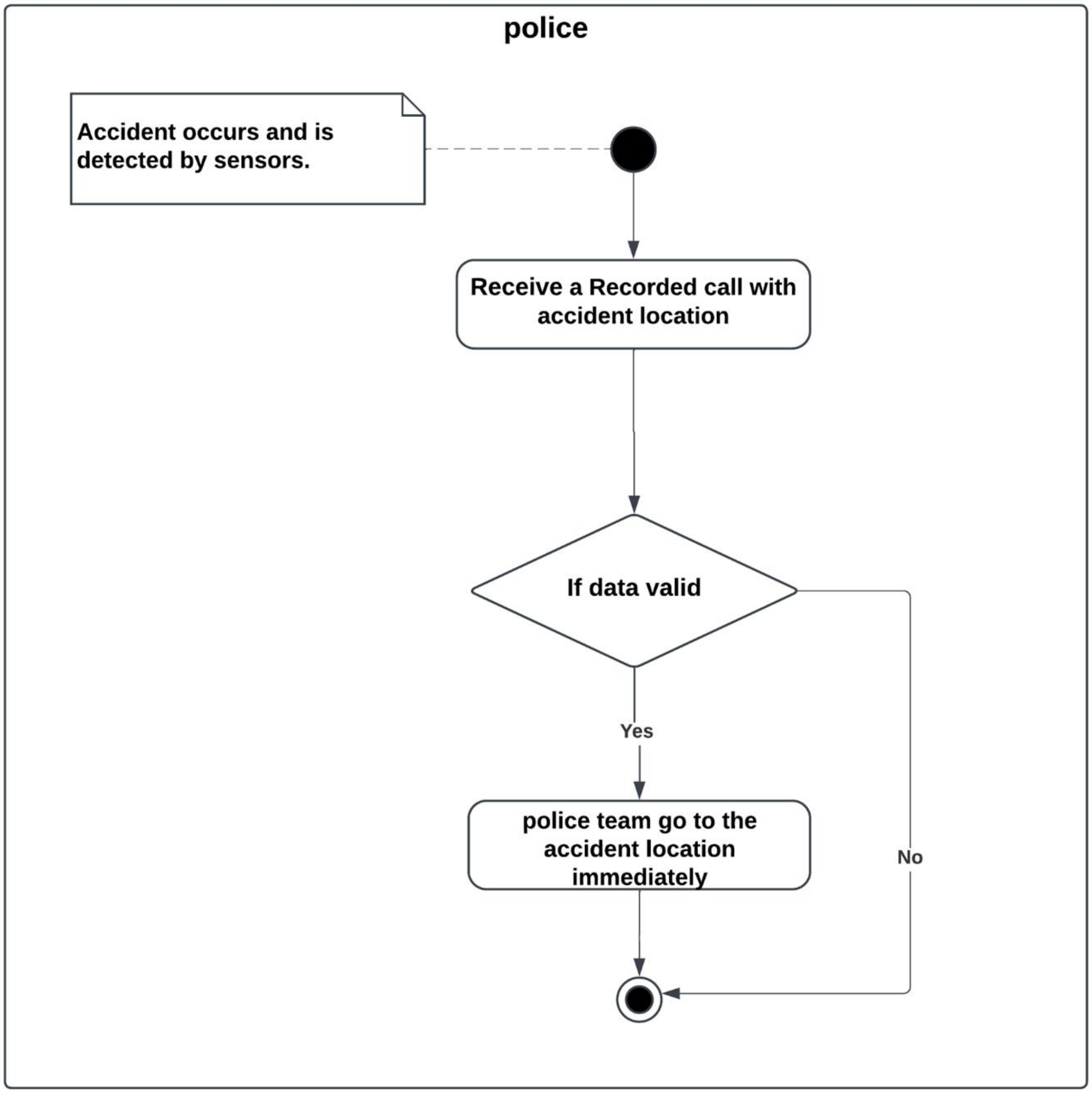
4.2.7: Finding Hospital



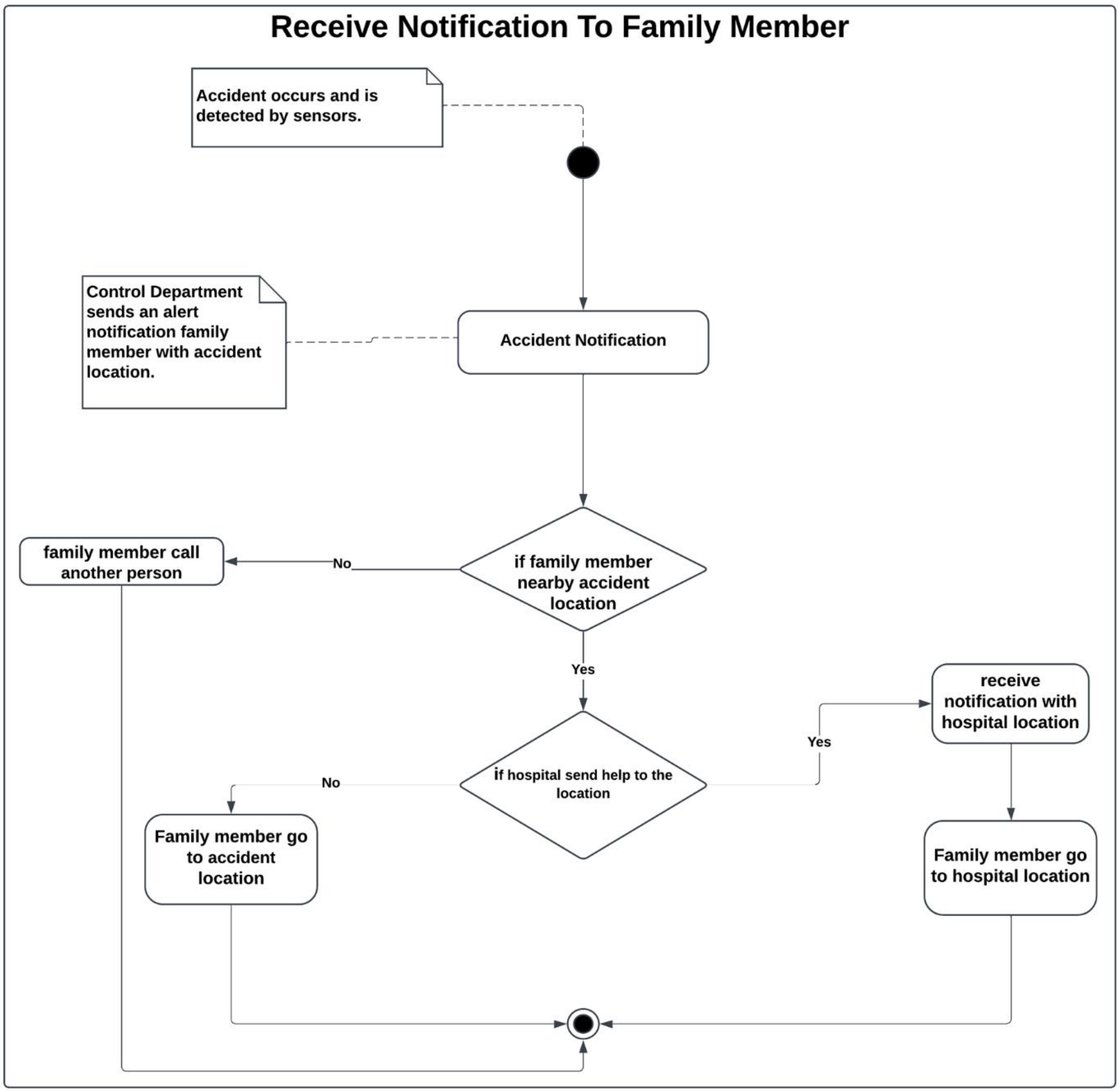
4.2.6: Control Department



4.2.8: Police

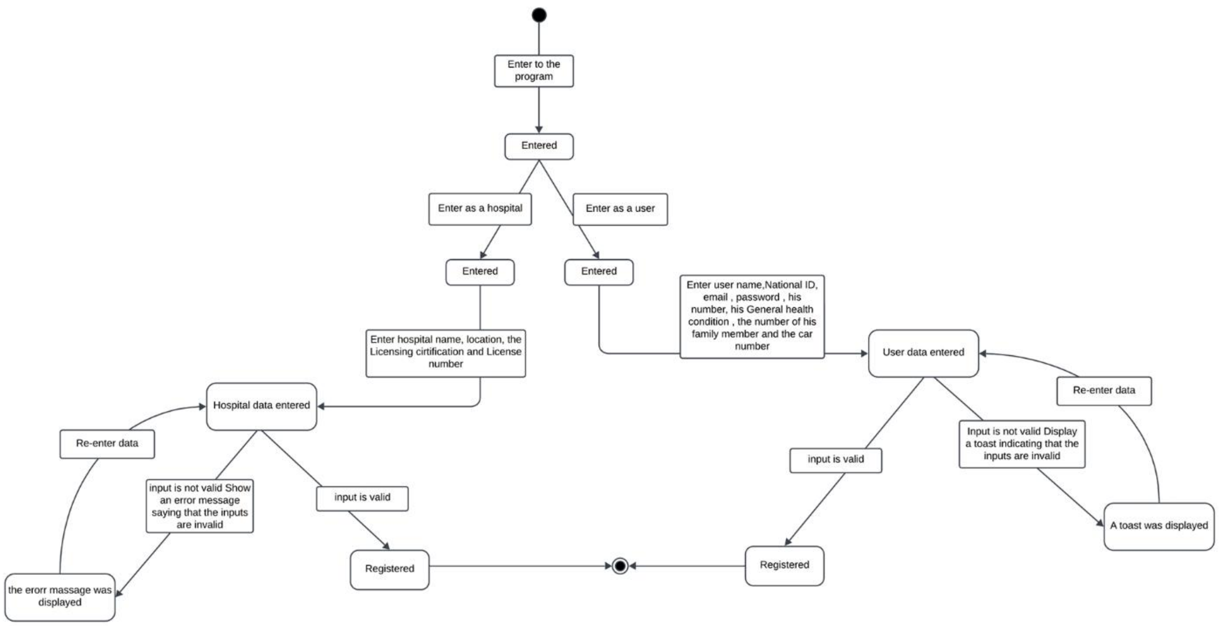

police
Accident occurs and is detected by sensors.
Receive a Recorded call with accident location
If data valid
Yes
No
police team go to the accident location immediately

4.2.9: Family Member

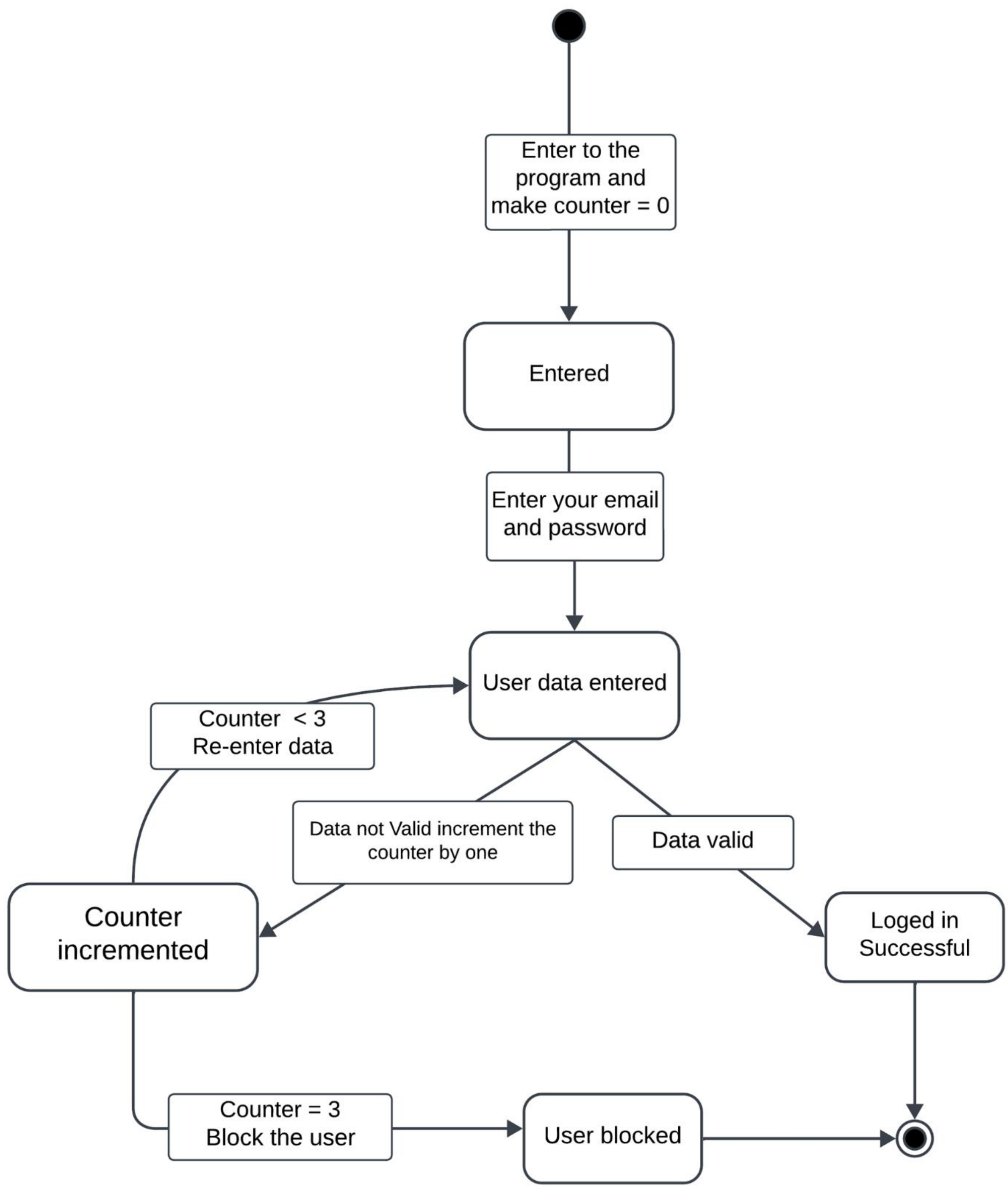


4.3 State diagram:

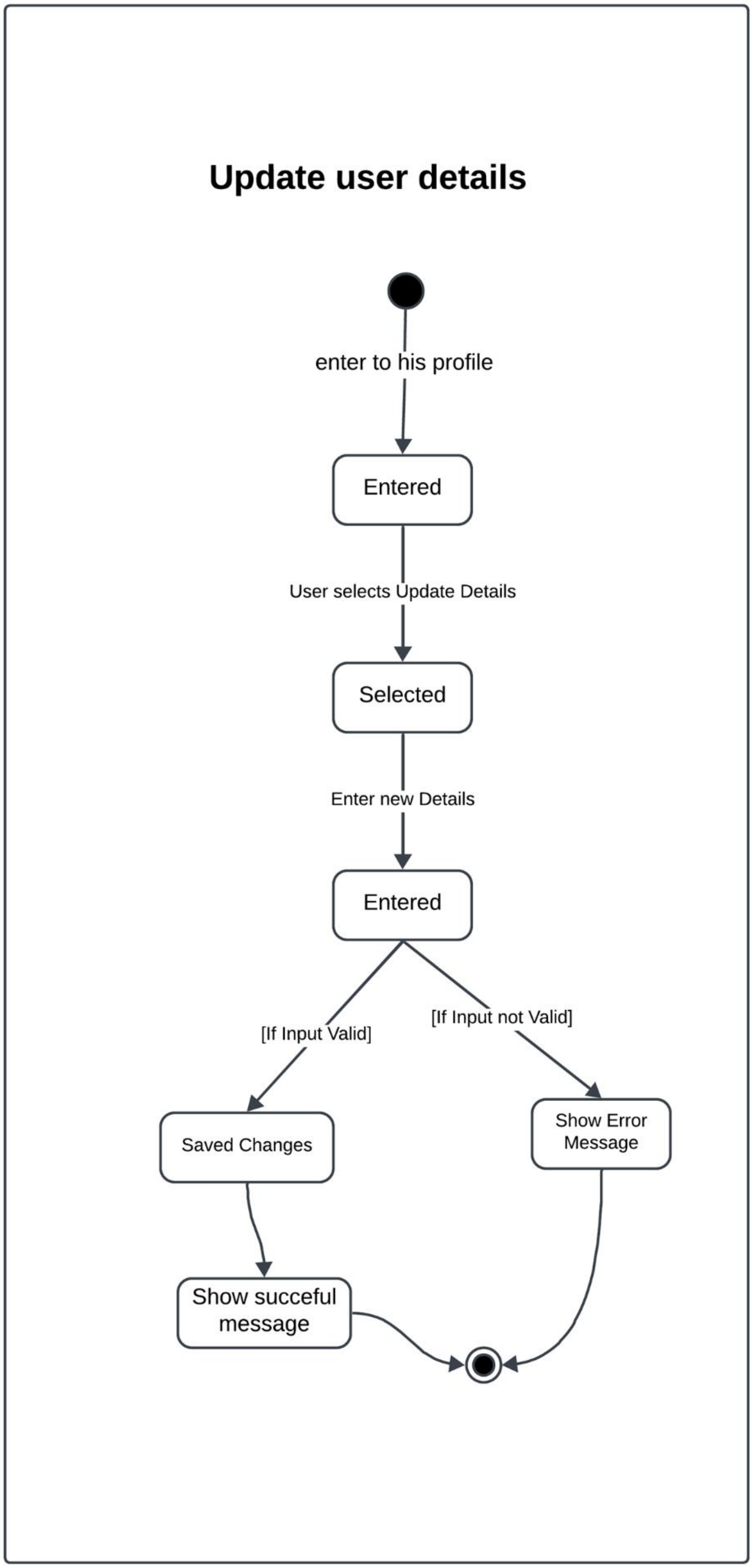
4.3.1: Register



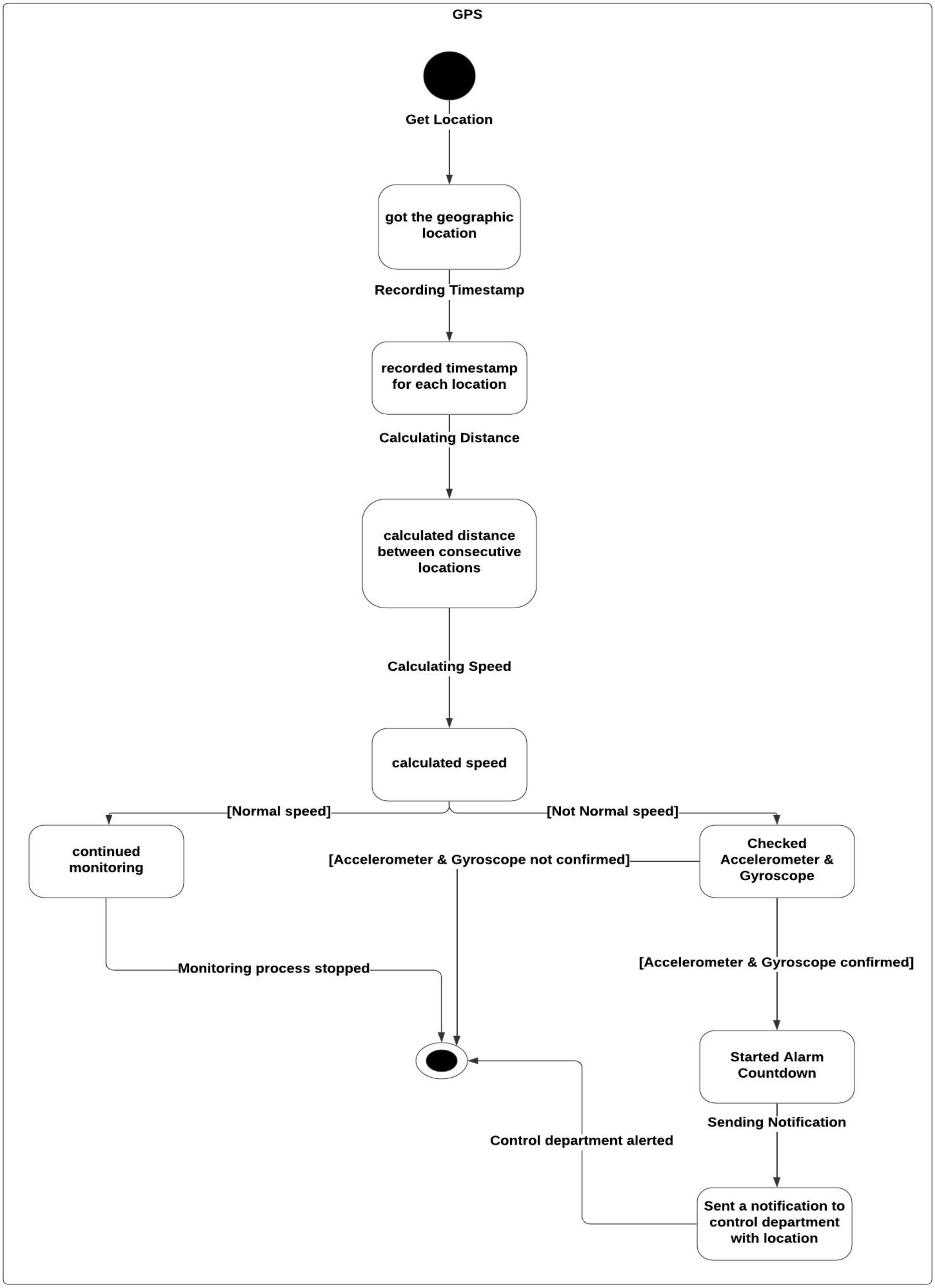
4.3.2: Login


Enter to the program and make counter = 0
Entered
Enter your email and password
User data entered
Counter < 3
Re-enter data
Data not Valid increment the counter by one
Data valid
Counter incremented
Loged in Successful
Counter = 3
Block the user
User blocked

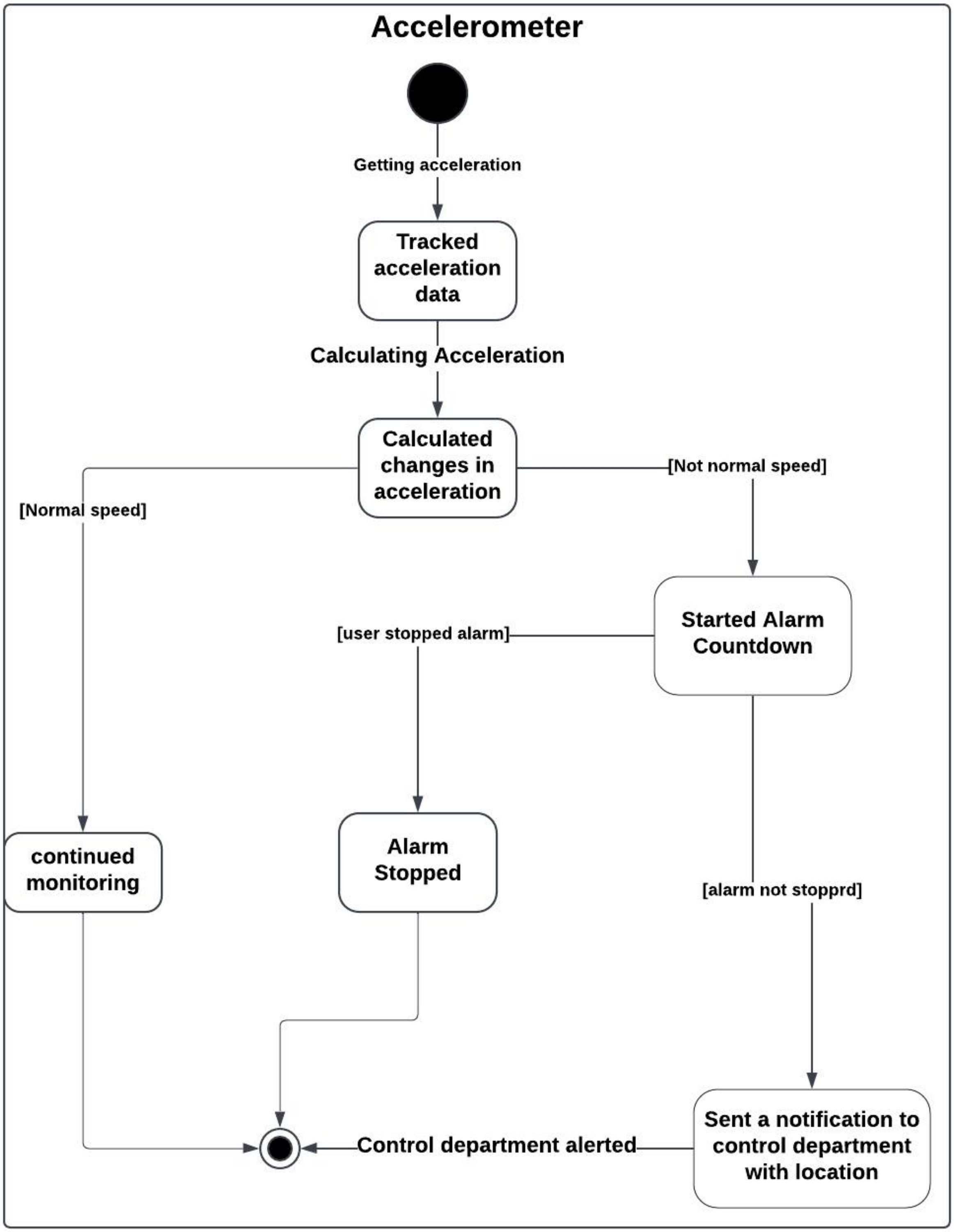
4.3.3: Update


Update user details
enter to his profile
Entered
User selects Update Details
Selected
Enter new Details
Entered
[If Input Valid]
[If Input not Valid]
Saved Changes
Show Error Message
Show succeful message

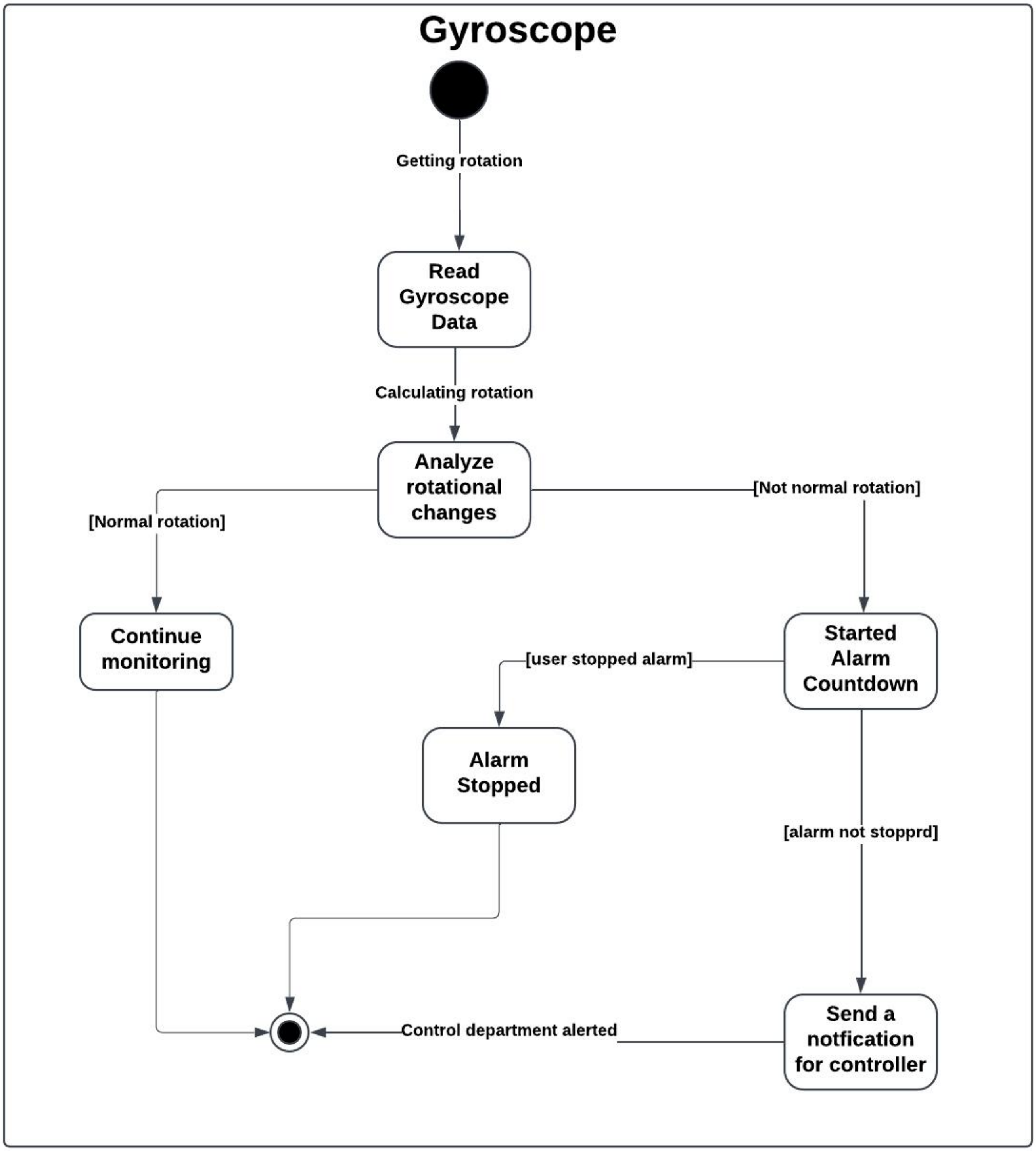
4.3.4: GPS


GPS
Get Location
got the geographic location
Recording Timestamp
recorded timestamp for each location
Calculating Distance
calculated distance between consecutive locations
Calculating Speed
calculated speed
[Normal speed]
[Not Normal speed]
continued monitoring
[Accelerometer & Gyroscope not confirmed]
Checked Accelerometer & Gyroscope
Monitoring process stopped
[Accelerometer & Gyroscope confirmed]
Started Alarm Countdown
Sending Notification
Control department alerted
Sent a notification to control department with location

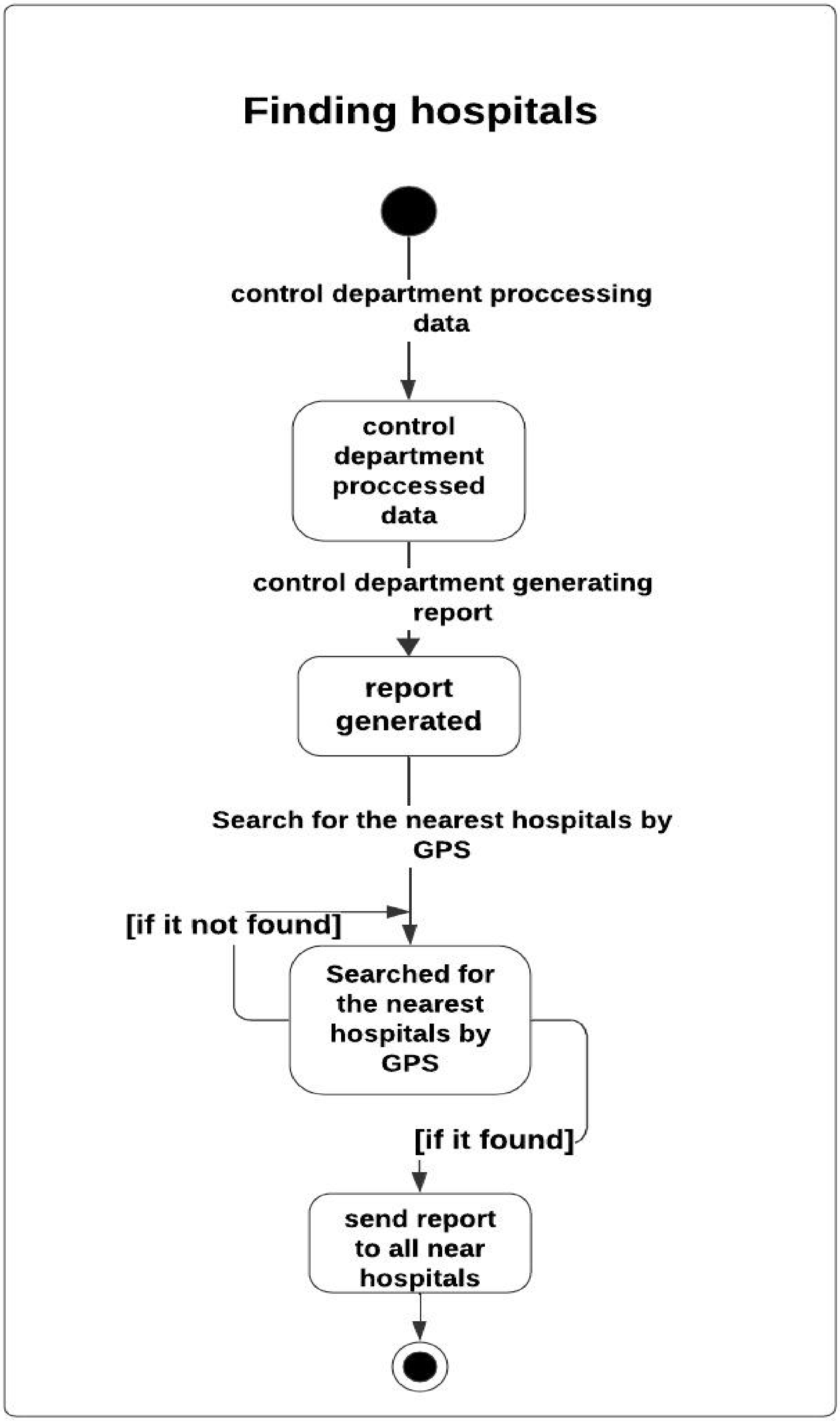
4.3.5: Acceleration



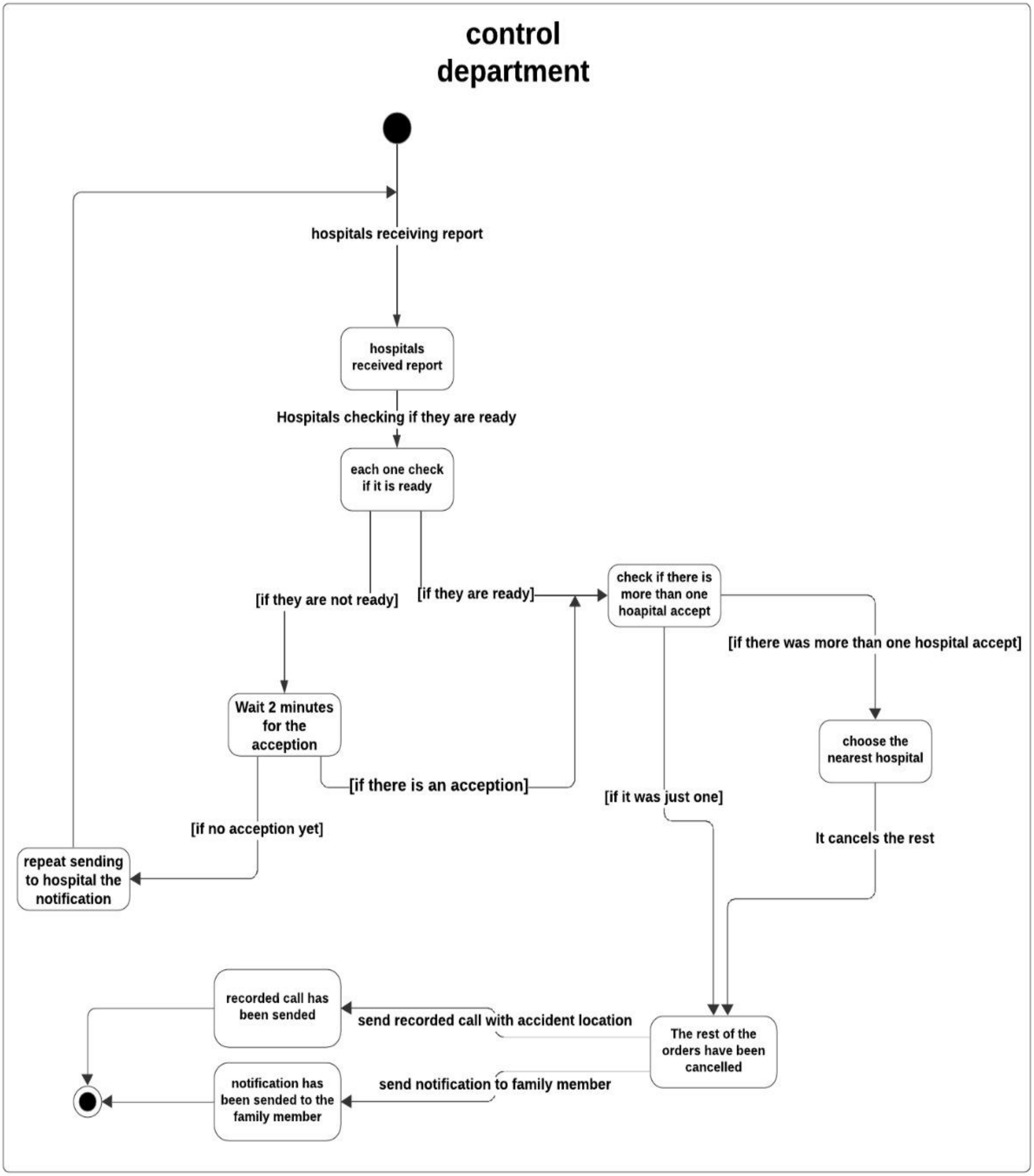
4.3.6: Gyroscope



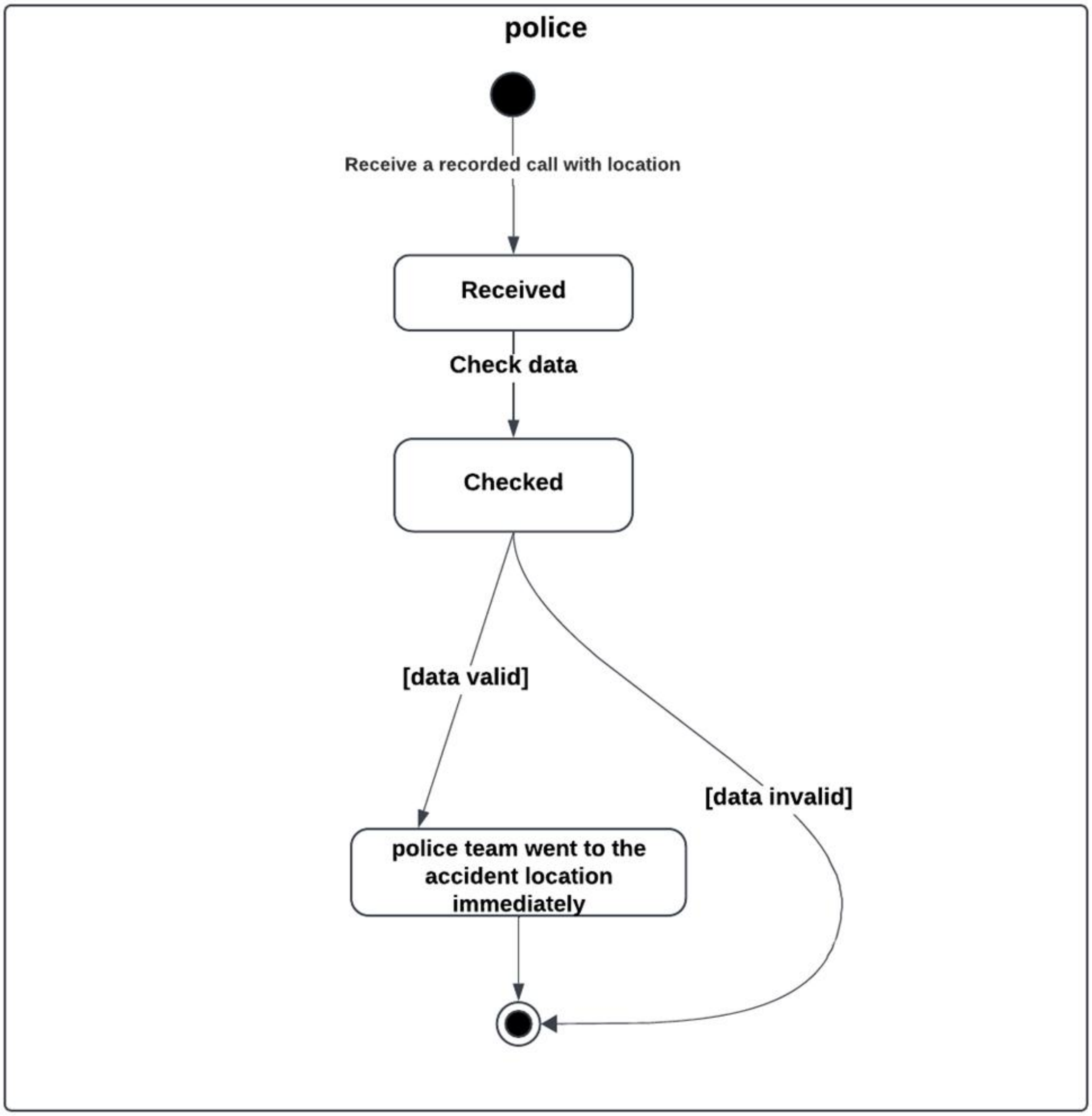
4.3.8: Finding Hospital



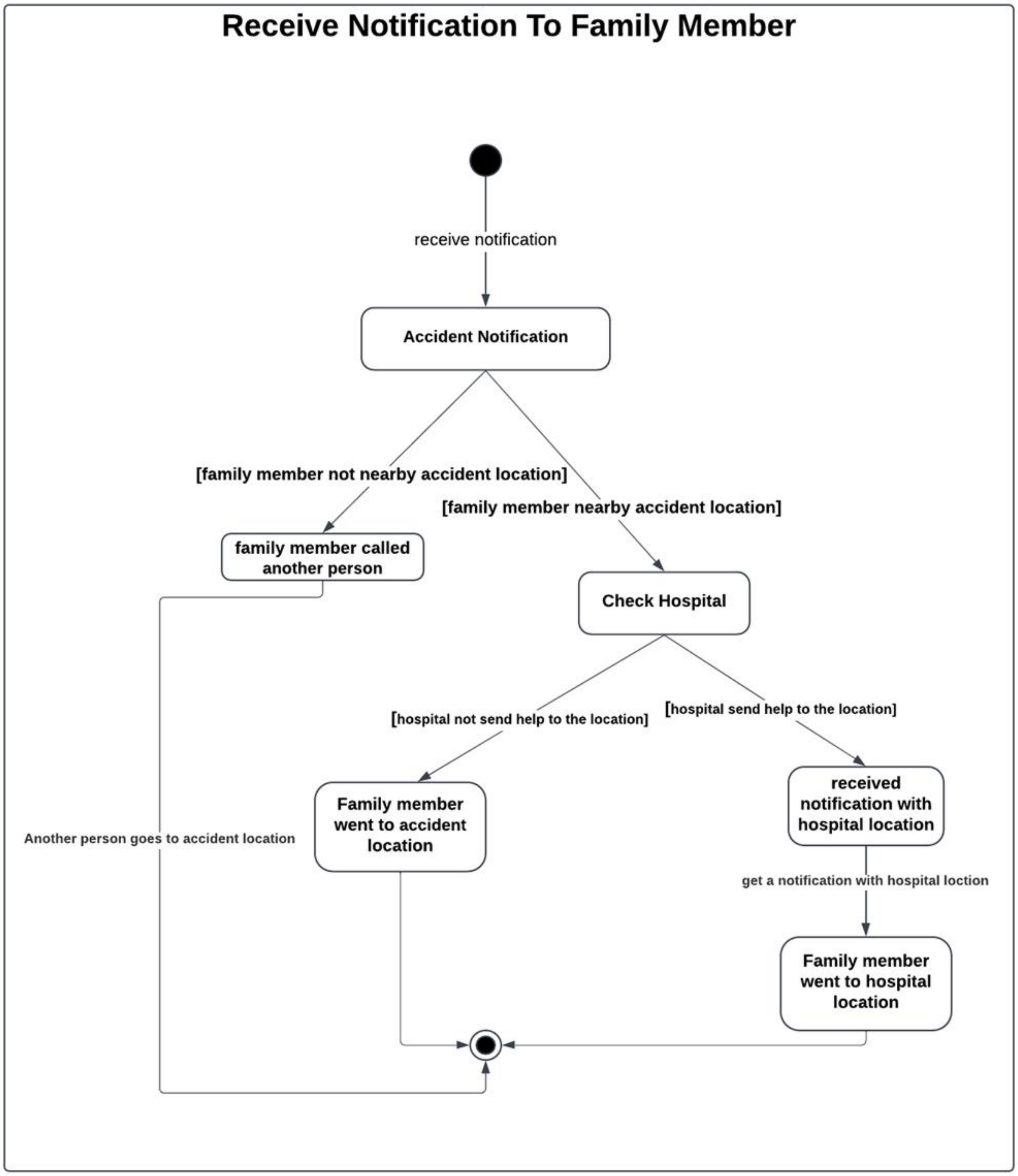
4.3.7: Control Department



4.3.9: Police

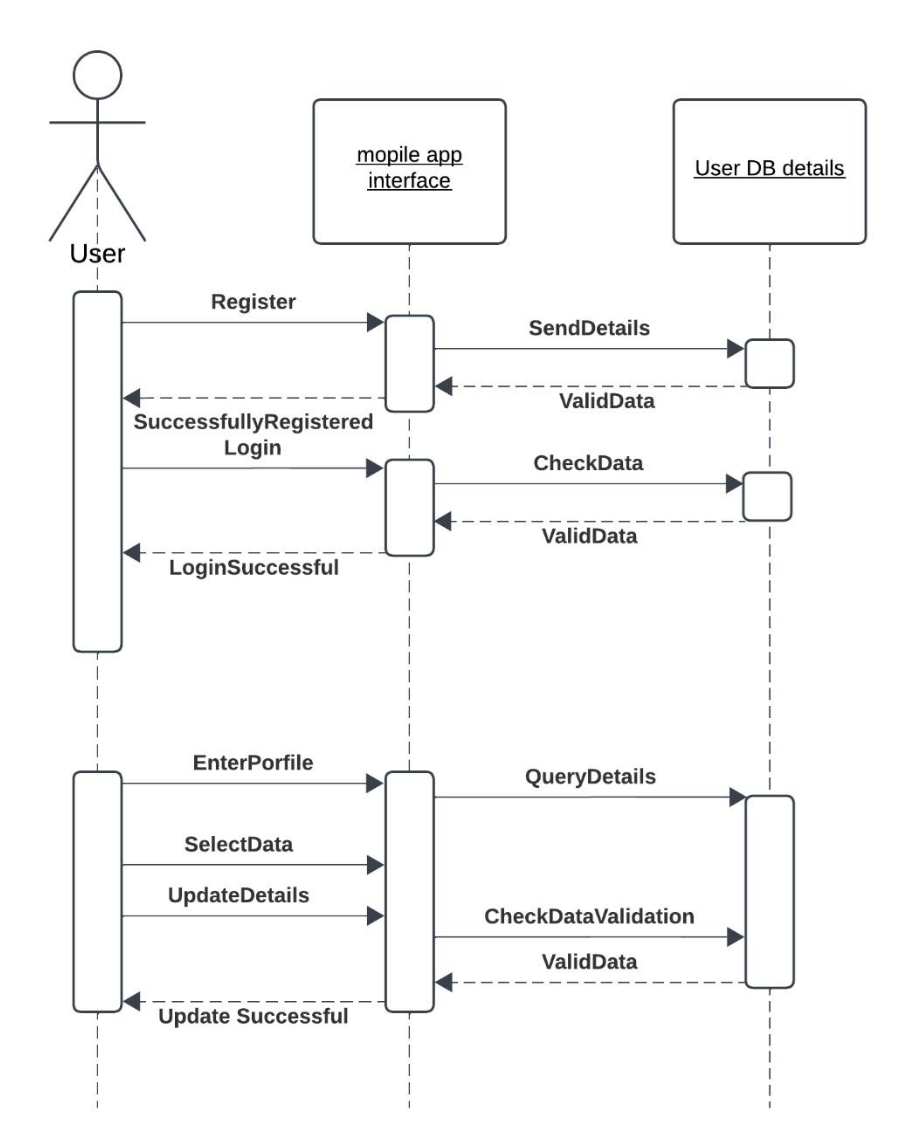

police
Receive a recorded call with location
Received
Check data
Checked
[data valid]
[data invalid]
police team went to the accident location immediately

4.3.10: Family Member

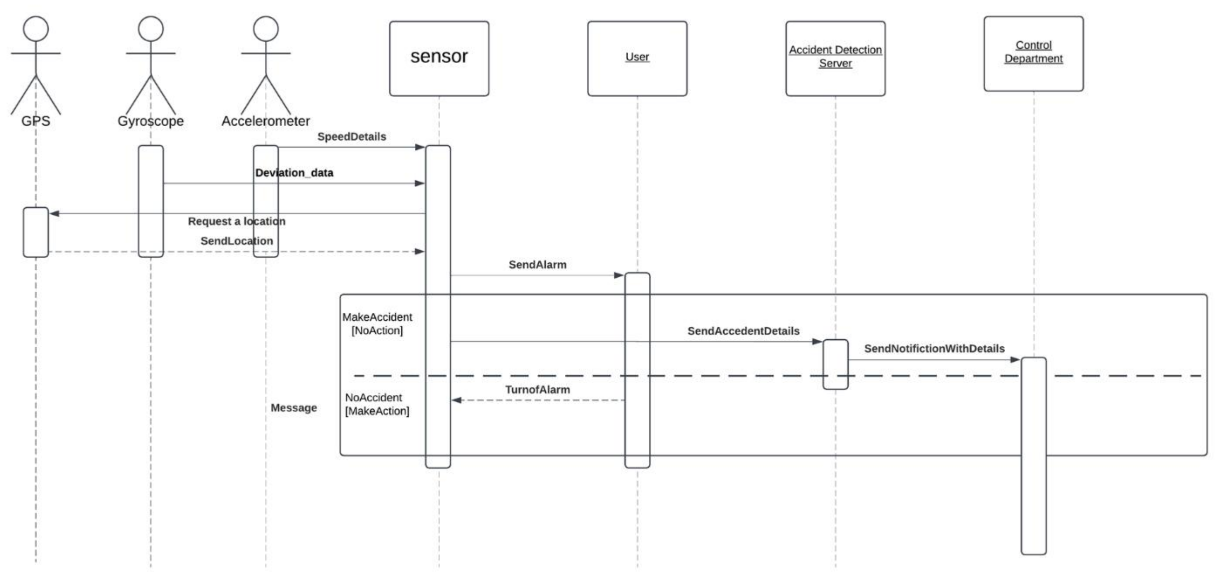


4.4 Sequence diagram:

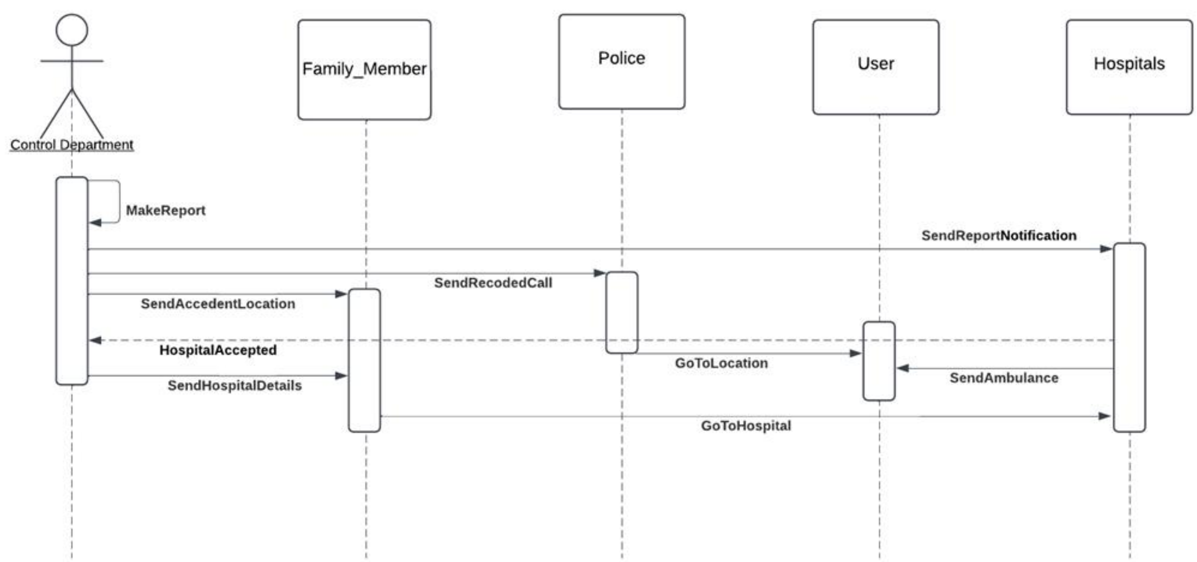
4.4.1: User



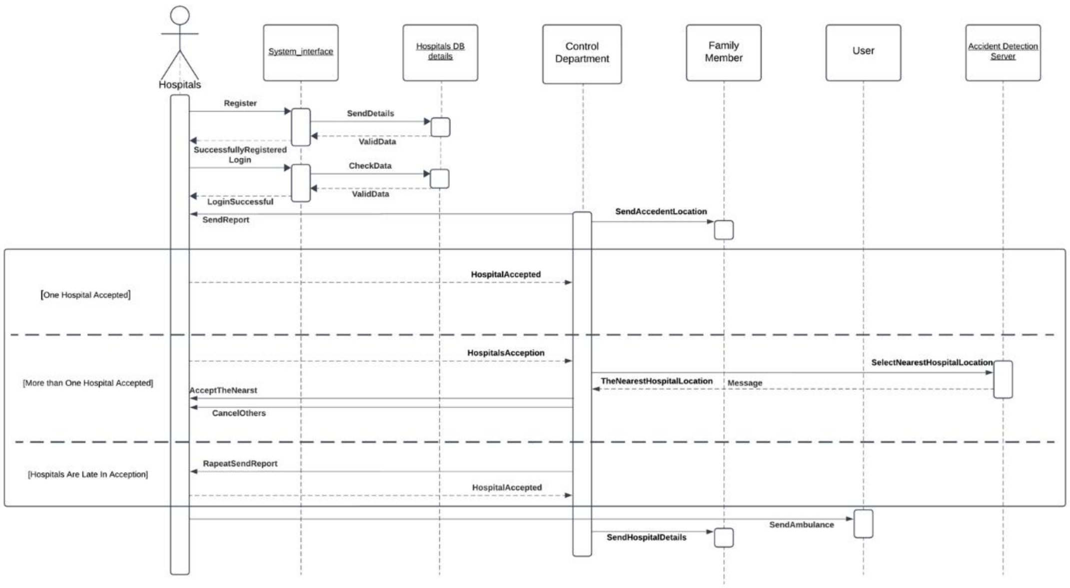
4.4.2: Sensors



4.4.3: Control Department

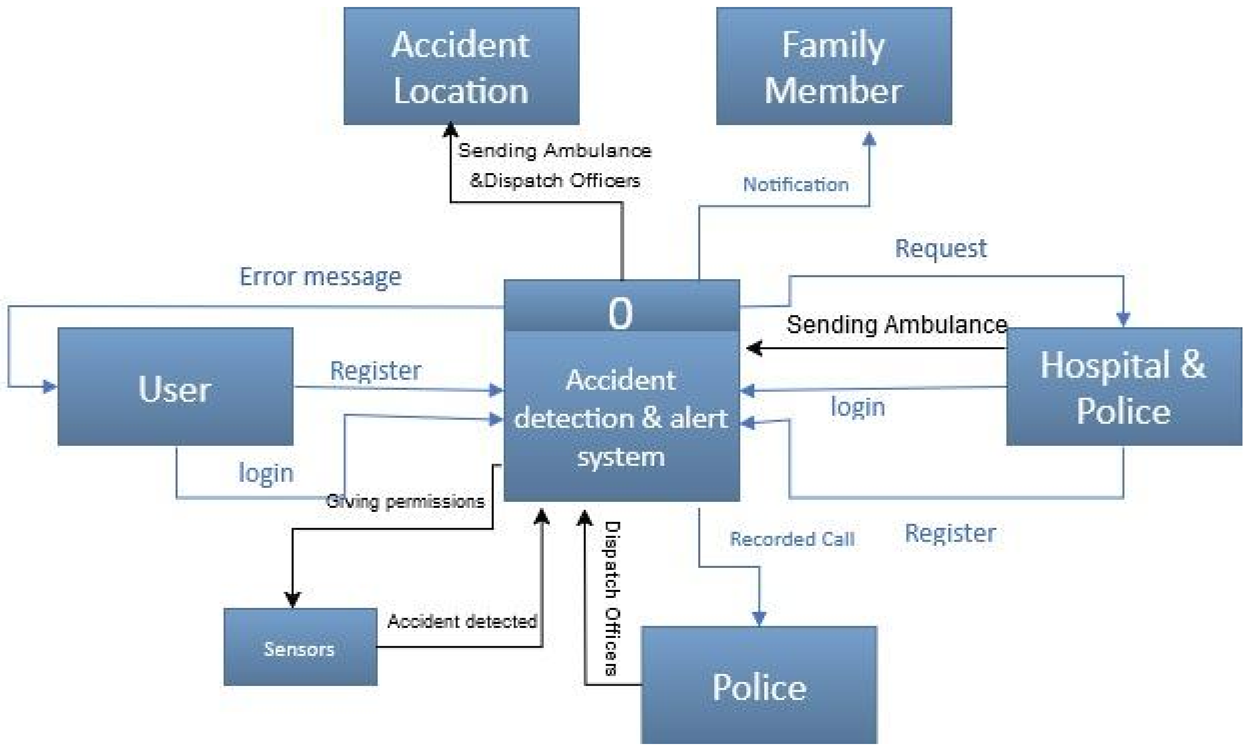


4.4.4: Hospital

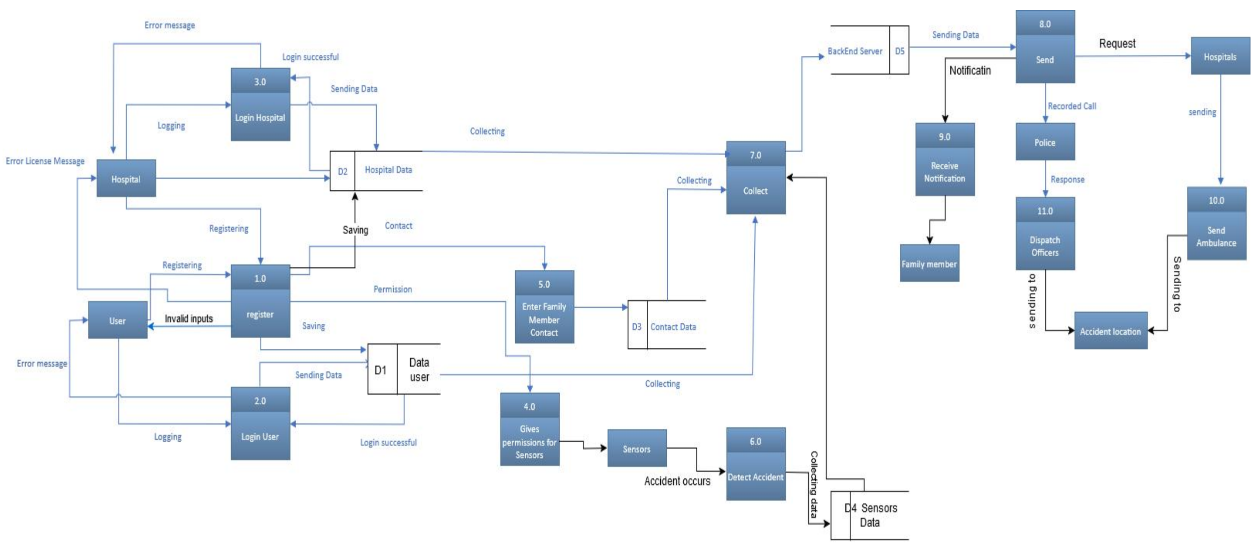


4.5 DFD:

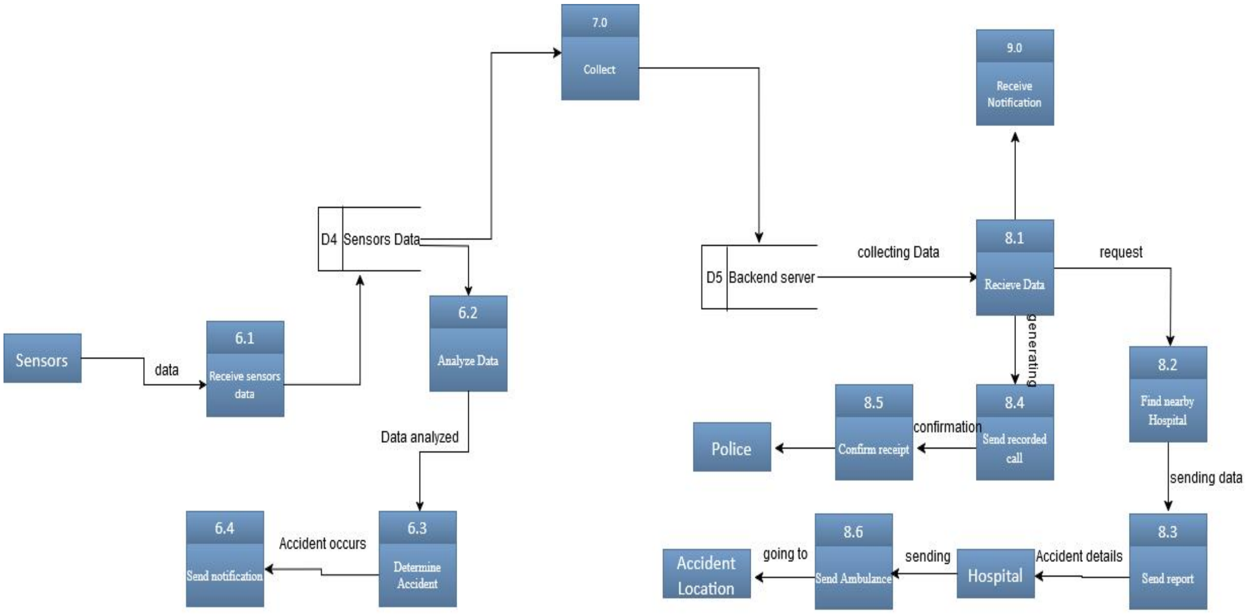
4.5.1: Context Diagram



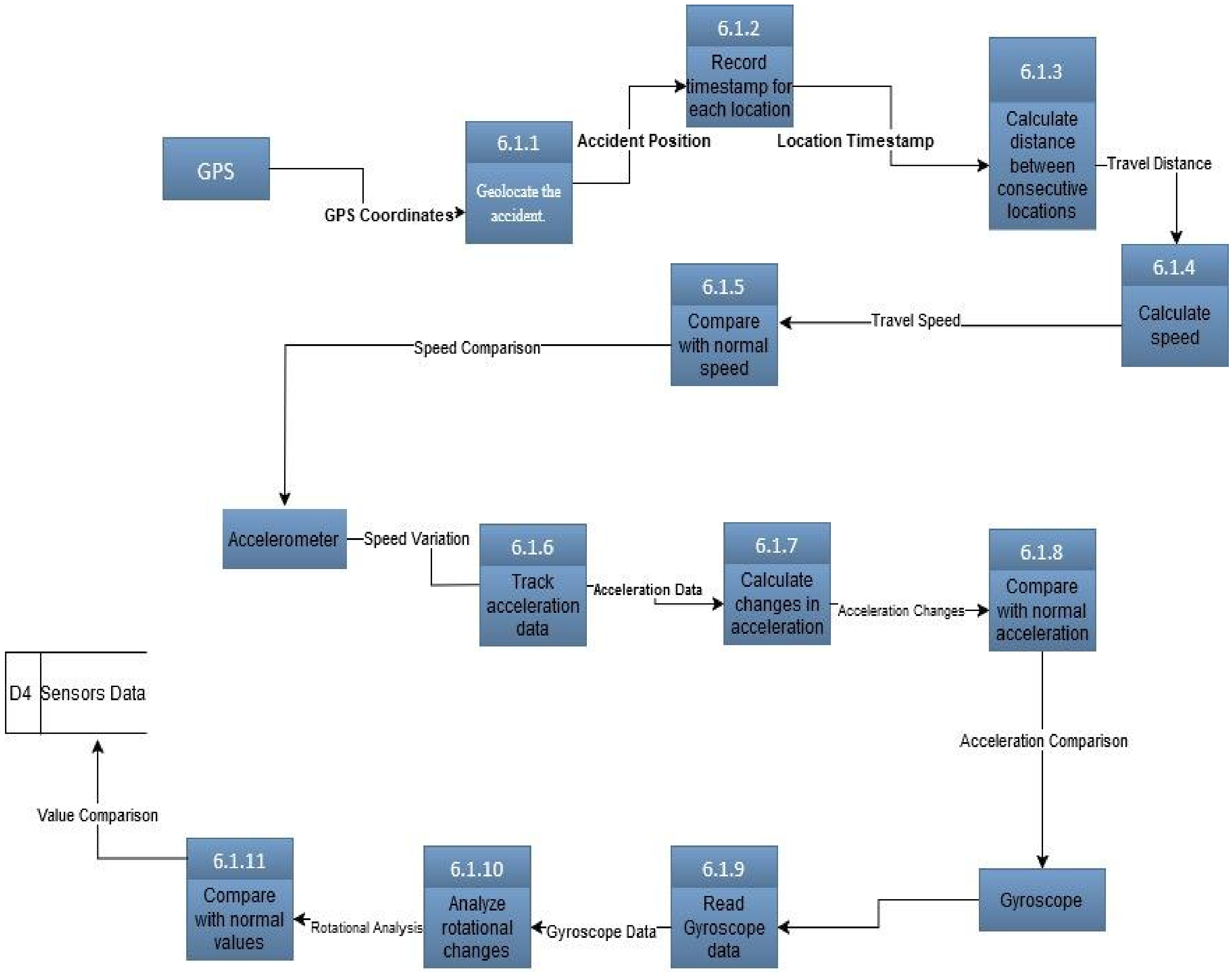
4.5.2: Level 0



4.5.3: Level 1


7.0
Collect
9.0
Receive Notification
D4 Sensors Data
D5 Backend server
collecting Data
8.1
Recieve Data
request
Sensors
data
6.1
Receive sensors data
6.2
Analyze Data
generating
8.2
Find nearby Hospital
8.5
Confirm receipt
confirmation
8.4
Send recorded call
Police
Data analyzed
sending data
6.4
Send notification
Accident occurs
6.3
Determine Accident
8.6
Send Ambulance
going to
Accident Location
sending
Hospital
Accident details
8.3
Send report

4.5.4: Level 2


6.1.2
Record timestamp for each location
6.1.3
Calculate distance between consecutive locations
GPS
6.1.1
Geolocate the accident
Accident Position
Location Timestamp
GPS Coordinates
Travel Distance
6.1.4
Calculate speed
6.1.5
Compare with normal speed
Travel Speed
Speed Comparison
Accelerometer
Speed Variation
6.1.6
Track acceleration data
Acceleration Data
6.1.7
Calculate changes in acceleration
Acceleration Changes
6.1.8
Compare with normal acceleration
D4 Sensors Data
Acceleration Comparison
Value Comparison
6.1.11
Compare with normal values
Rotational Analysis
6.1.10
Analyze rotational changes
Gyroscope Data
6.1.9
Read Gyroscope data
Gyroscope